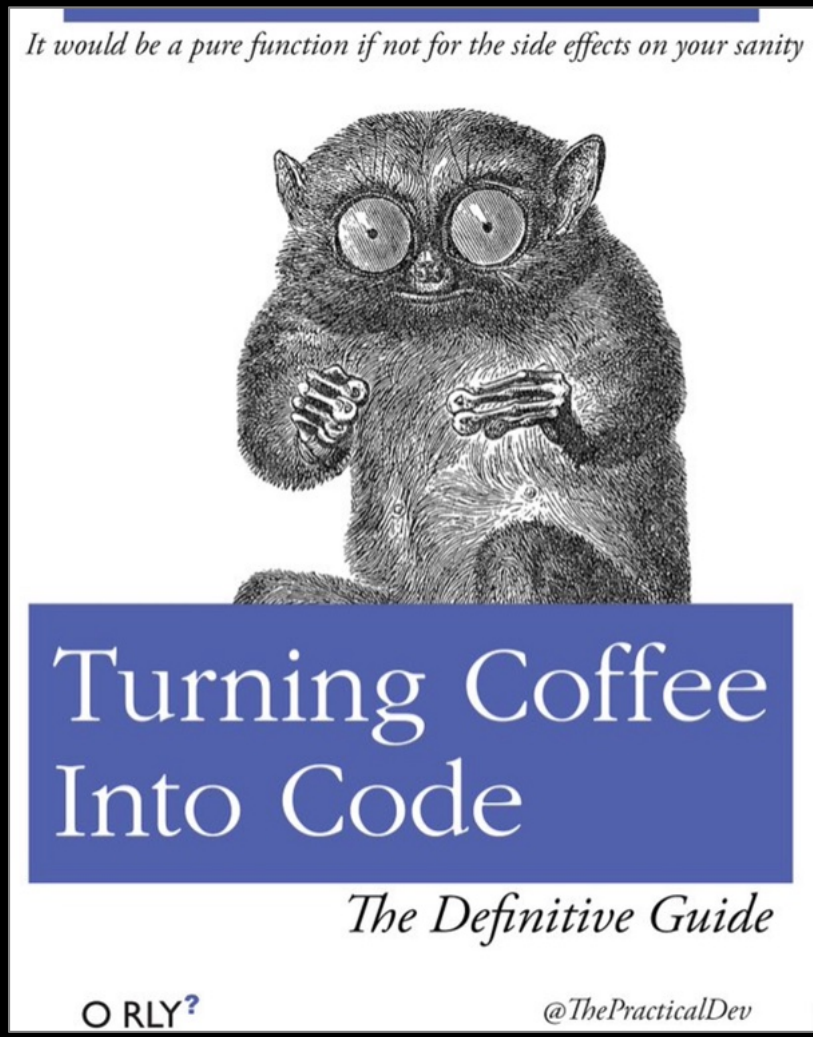
Web Technologies

Web programming (IV)

Web application servers

PHP language and environment



Prof. Sabin Corneliu Buraga – profs.info.uaic.ro/sabin.buraga/

Assoc. Prof. Andrei Panu – profs.info.uaic.ro/andrei.panu/

“Poor is the pupil
who does not surpass his master.”

Leonardo da Vinci

(re)discover...

 how do we use a Web application server

 essential information about PHP

 PHP as a procedural programming language

 object-oriented programming in PHP

 PHP features regarding Web interaction

 how can databases be accessed in PHP

 useful tools for Web developers

 case studies of Web applications developed in PHP

How do we use an application server for developing a Web application?

web application server

Purpose:
increasing the efficiency of all processes involved in the development of Web applications

web application server

Integrated in one/many Web servers

can also provide its own Web server
or runtime environment

web application server

Can encourage or enforce an architectural vision regarding Web application development

typical situation:
MVC or variants

web application server

Simplifies the invocation of programs (scripts) concerning a Web application

- dynamic content generation on the server side

web application server

Aspects of interest:

programming language(s)
core API
persistent storage of data models
Web interaction
cookies and sessions
development environments + frameworks, components,…
specific characteristics

web application server

Persistent storage

in relational databases – using SQL

web application server

Persistent storage

in relational databases – using SQL

examples:
ADO.NET for ASP.NET
Java – JDBC (Java DataBase Connectivity)
PHP – predefined functions/modules, plus included libraries (SQLite + mysqli) or various extensions

advanced

web application server

Persistent storage

in relational databases – using SQL

ORM (Object-Relational Mapping): Data Mapper pattern

Go – Xorm framework
Java – JPA (Java Persistence API) specification
+ implementations: EclipseLink, Hibernate, OpenJPA,…
Node.js – Sequelize library
PHP – frameworks: Cycle ORM, Doctrine, Propel, RedBean, etc.

advanced

web application server

Persistent storage

in relational databases – using SQL

Active Record - architectural pattern used in ORM to encapsulate a table or a view into a class

examples:
active_record, TypeORM (Node.js modules), Castle Project (.NET), DBIx::Class (Perl), Django ORM and Orator (Python), Play Framework (Java, Scala), Rails (Ruby)

web application server

Persistent storage

tree-based models: XML

(semi)structured data
transformations in other formats: XPath, XSLT
processing: DOM, SAX, SimpleXML, etc.
data validation: DTD, XML Schema, RELAX,…
querying: XQuery

advanced

web application server

Persistent storage

using other non-relational paradigms
(based on graphs and/or key-value)

distributed on Internet, scalable – NoSQL

github.com/erictleung/awesome-nosql-guides

dbdb.io • github.com/igorbarinov/awesome-data-engineering#databases

examples:

Cassandra, MarkLogic, MongoDB, Neo4j, OpenLink Virtuoso, Redis

web application server

Web interaction

facilitated by specific controls specified in the source code invoked on the server

web application server

Web interaction

facilitated by specific controls specified in the source code invoked on the server

can emulate HTML form fields and/or provide new interactive controls – e.g., calendar, slideshow,…

- generation of processable code at client level (front-end)
  Web components (HTML + CSS + JavaScript) executed by the browser

advanced

web application server

Web interaction

examples:

ASP.NET (**<asp:control>** – e.g., FileUpload, ListBox, Table,...)

PRADO framework (PHP)

formidable, form-data, forms – Node.js modules

Java platform: JSF (Jakarta Server Faces)

web application server

Web interaction

encourages the use of templates
based on a specific processor

Web template system

web application server

Web interaction

Web template system

using specifications of content presentation
(a Web template), persistent data (e.g., retrieved from
a database) is used by a processor (template engine)
in order to generate HTML documents
or representations in other formats

```
<!-- HTML template -->
<h1 class="profile">[@username] profile</h1>
<img src="[@photoURL]" alt="photo"/>
<div class="identity">[@firstName] [@lastName]</div>
<div class="location">[@location]</div>

<!-- PHP program: template processing -->
$profile = new Template ('templates/profile.tpl');
$profile->set ('username') = 'Tux';
$profile->set ('photoURL') = 'imgs/tux.svg';
$profile->set ('firstName') = 'Tuxy';
$profile->set ('lastName') = 'Pinguinesscool';
$profile->set ('location') = 'Romania';
```

Tux profile

Tuxy Pinguinesscool
Romania

example:
the specification of content presentation
(HTML template – **.tpl**) includes variable names
– here, using **[@*variable*]** syntax – that will be substituted by
actual values obtained from the program as a result
of the Web template system invocation

web application server

Web interaction

Web template system

on the server

Apache FreeMarker (Java), Blade (PHP), Haml (Ruby), Mustache (C++, JS, PHP, Python, Scala,…), Pug (Node.js), Razor (.NET), Smarty (PHP), Tonic (PHP), XSLT (XML)

web application server

Web interaction

Web template system

on the client

available for JavaScript:
HandleBars, Mustache.js, Nunjucks,...
github.com/sorrycc/awesome-javascript#templating-engines

advanced

web application server

Web interaction

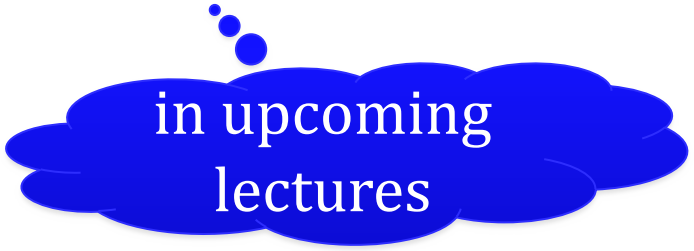
cookie and session management

via data structures/types – examples:
HttpSession class (ASP.NET), HttpSession interface (Java servlets), HTTP::Session (Perl), session (Flask – Python framework), web.session (web.py), HttpFoundation (Symfony component – PHP framework), SessionComponent class (CakePHP), session array (Ruby on Rails), play.mvc.Http.Cookie (Play for Java/Scala), sessions (Gorilla – Go) cookie-parser and express-session (Node.js modules for Express)

web application server

Web interaction

asynchronous data transfer via Ajax technology suite



eventually, via additional frameworks/modules/classes

advanced

web application server

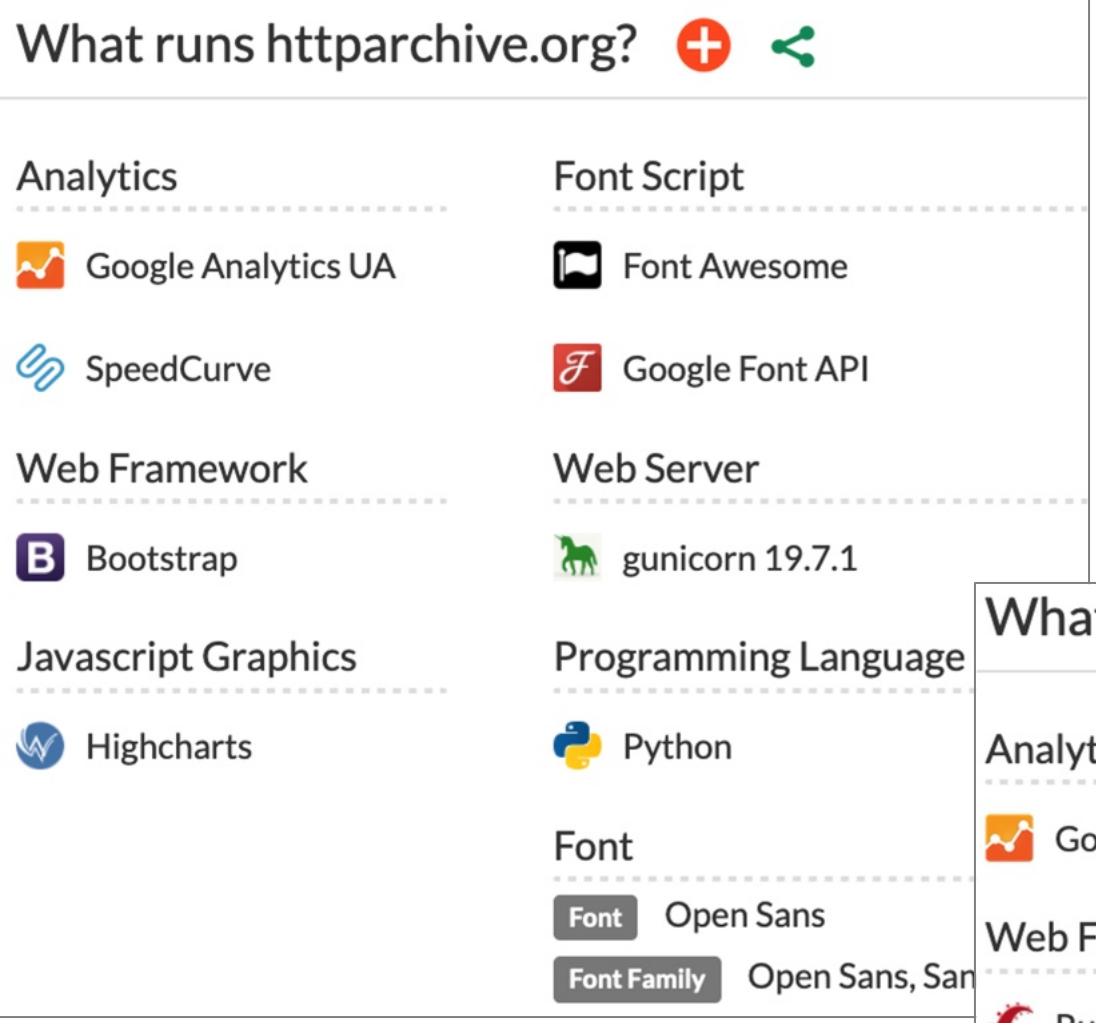
Support for software engineering

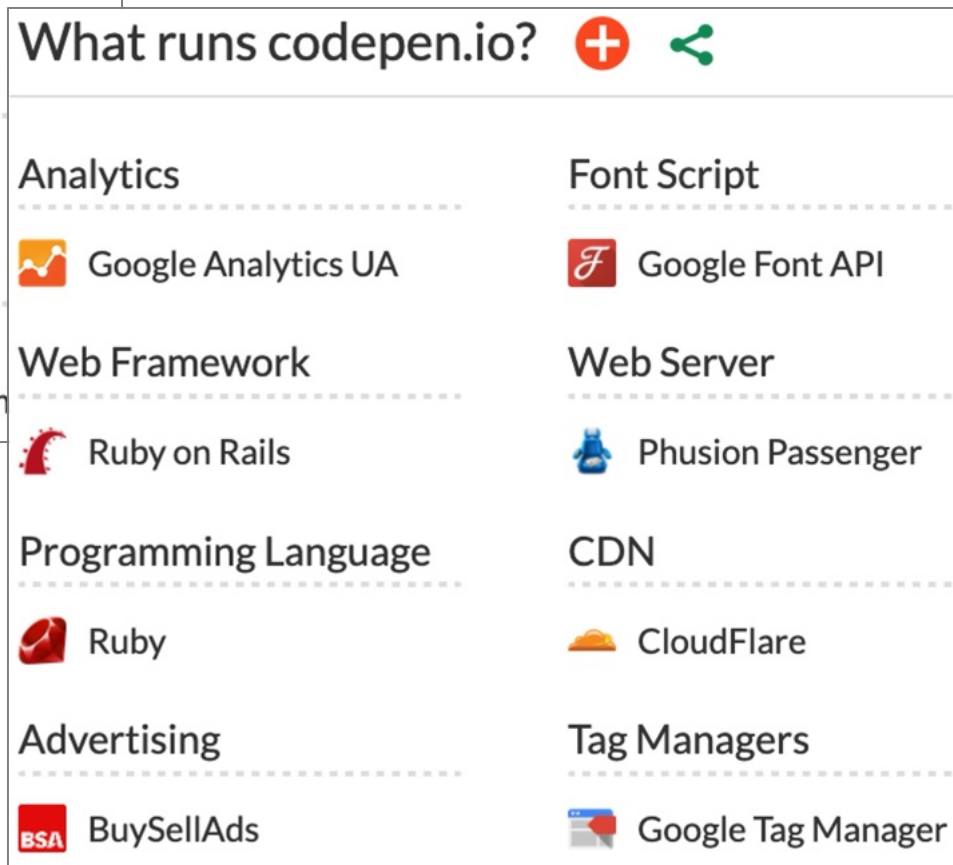
encouraging/enforcing the use of **design patterns** (arhitectural, behavioral, structural, targeting concurrent and/or distributed processing and others)

AMI (asynchronous method invocation), broker, CBD (components-based development), DAO (data access object), DDD (domain-driven design), DTO (data transfer object), façade, MOM (message-oriented middleware), microservice, **MV***, publish-subscribe, SOA (service-oriented architecture), singleton,…

example: *Spaghetti Code* **anti-pattern**

usually, specific to Web applications that mix business logic with content presentation (view) and data model access mechanism





inspecting the technologies used by a Web application with the instruments Wapplyzer and WhatRuns

web application server

Integrated into a software stack

software stack (servers, tools,…) offering support for developing complex Web applications

web application server

Integrated into a software stack

software stack (servers, tools,…) offering support
for developing complex Web applications

available – usually, *open source* –
for a specific platform
(operating system, Web server, database server,
application server, programming language)

web application server

Integrated into a software stack

LAMP
(Linux, Apache HTTP Server, MariaDB/MongoDB, Perl/PHP/Python)

alternatives:
FAMP (FreeBSD), MAMP (macOS),
WAMP (Windows), XAMPP (multi-platform)

www.apachefriends.org

advanced

- Adminer
- Apache
- Drupal
- Joomla
- Laravel
- Mariadb
- Memcached
- Mongodb
- Mysql
- Nginx
- Php
- Phpmyadmin
- Postgresql
- Redis
- Wildthing
- Wordpress

wamp.net

tools for
Web developers

advanced

web application server

Integrated into a software stack

based on JavaScript – full stack Web development

MEAN (MongoDB, Express, Angular, Node.js)

MERN (MongoDB, Express, React, Node.js)

advanced

web application server

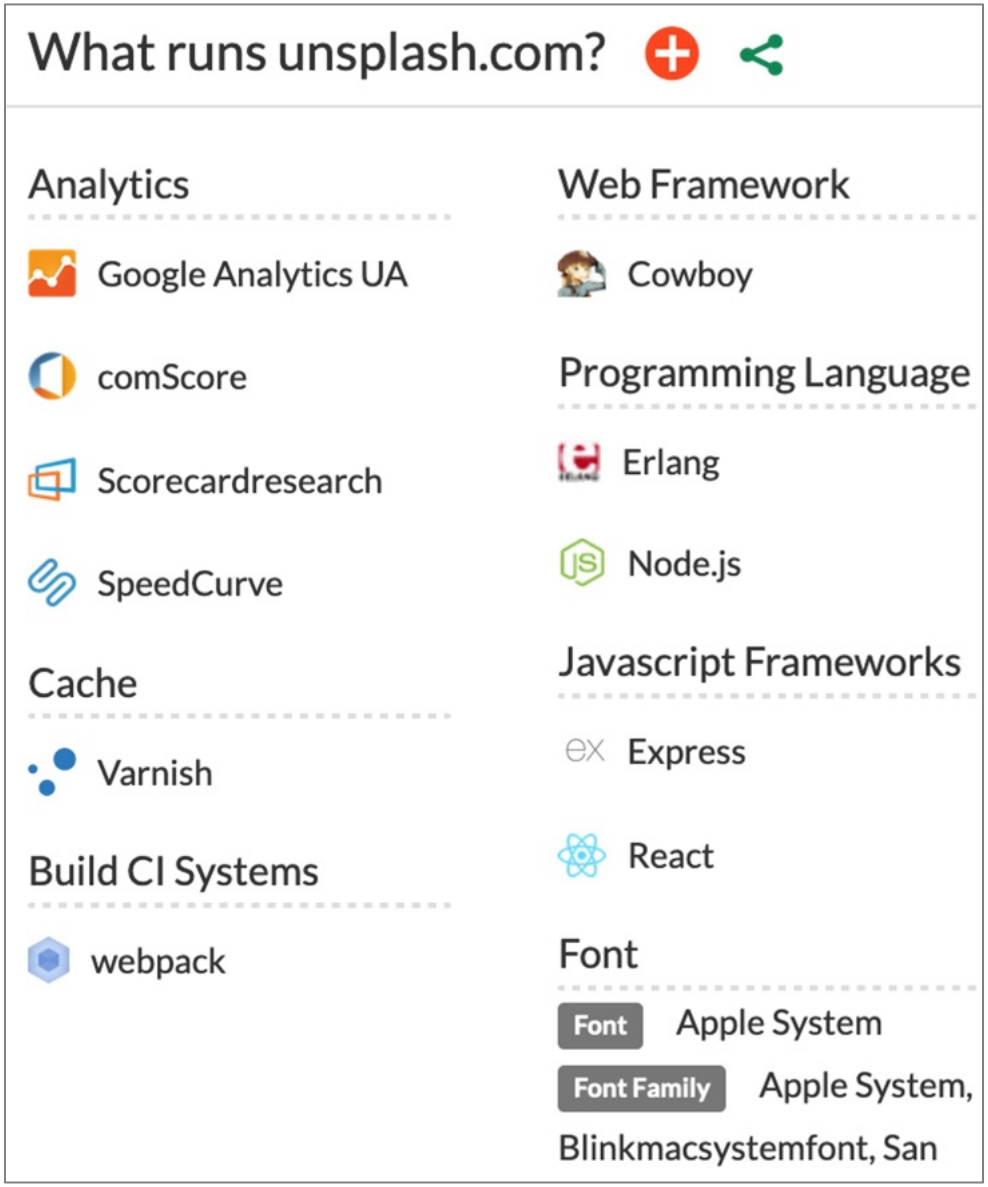
Integrated into a software stack

complementary approaches:

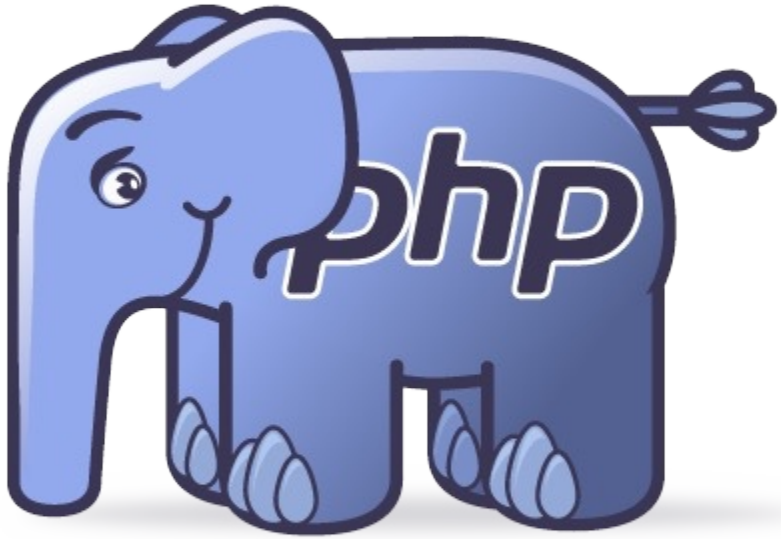
LAPP (Linux, Apache, PostgreSQL, Perl/PHP/Python)

TALL (Tailwind CSS, Alpine.js, Laravel, Livewire)

tallstack.dev



inspecting the technologies used by a Web application with the instrument WhatRuns



🐘 Essential information about PHP?

php

History

Important characteristics

PHP as programming language

paradigms: procedural, object-oriented, functional

PHP as Web development platform

interaction, database access, frameworks, libraries and tools, case studies

Personal Home Page Tools (1995)
Rasmus Lerdorf

PHP 3 (1998)
developed by Zend – Zeev Suraski & Andi Gutmans

PHP 4 (2000)
support for object-oriented programming

PHP 5 (2004) – most recent version: **PHP 5.6** (2014)
new features inspired by Java

PHP 7 (2015), **PHP 7.2** (2017), **PHP 7.4** (2019)
strong typing, support for Unicode, performance,…

PHP 8 (2020), **PHP 8.4** (2024) , **PHP 8.5** (2025)
major update: real-time compilation, new data types and syntactic constructs, improvements, etc.

php: characteristics

Web application server

provides a script-based interpreted programming language

can be directly included into HTML documents

php: characteristics

PHP is a procedural language, offering support for other programming paradigms
(object-oriented and, more recently, functional)

can be used as a general purpose language, too

reference book – freely available on the Web:
Josh Lockhart, *PHP: The Right Way*, 2026
phptherightway.com

php: characteristics

Syntax inspired by C, Perl and Java – case sensitive

whitespaces (space character, Tab, New Line) have no effect on the execution of the program

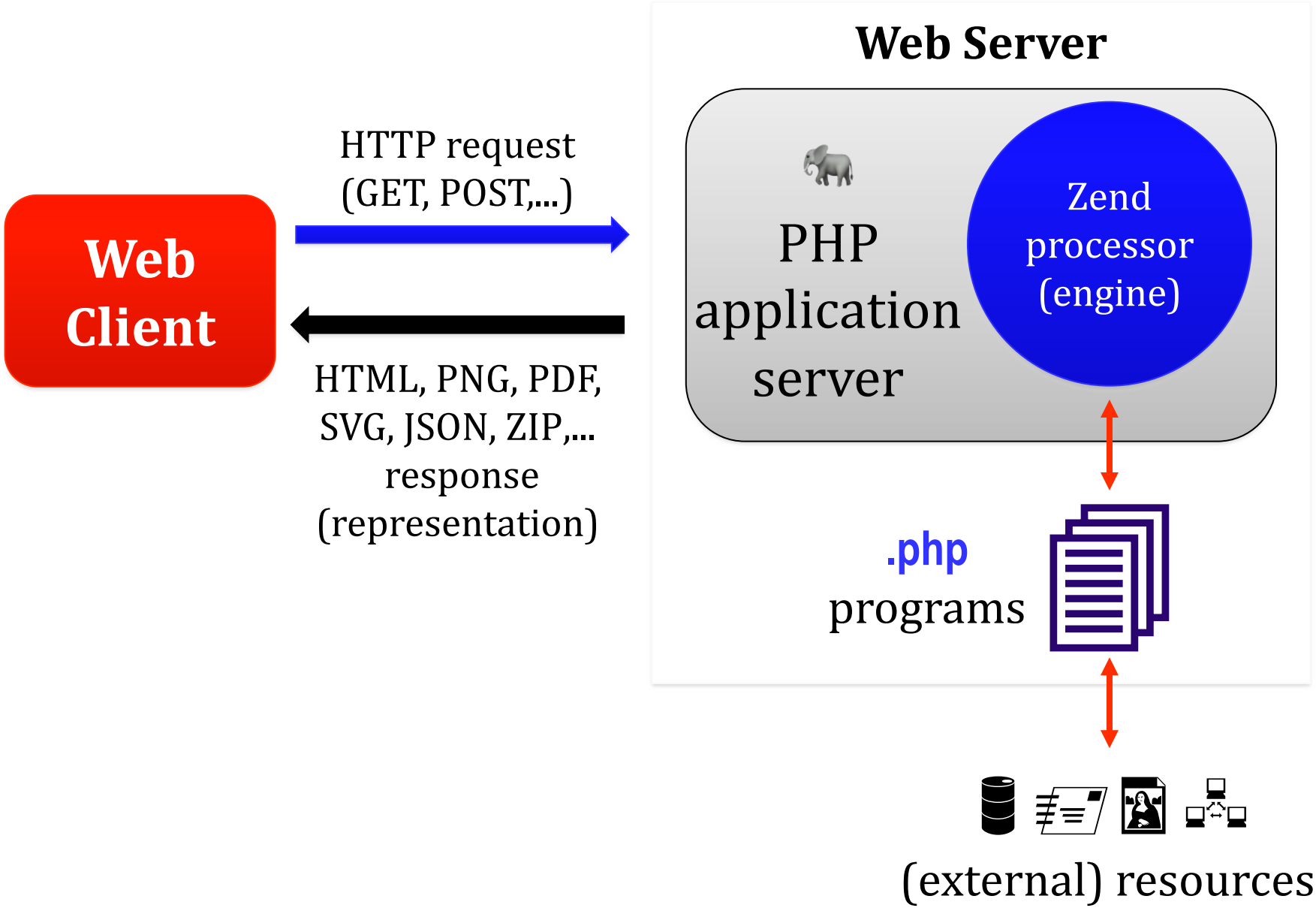
usually, the files which contain PHP source code have the extension **.php**

php: characteristics

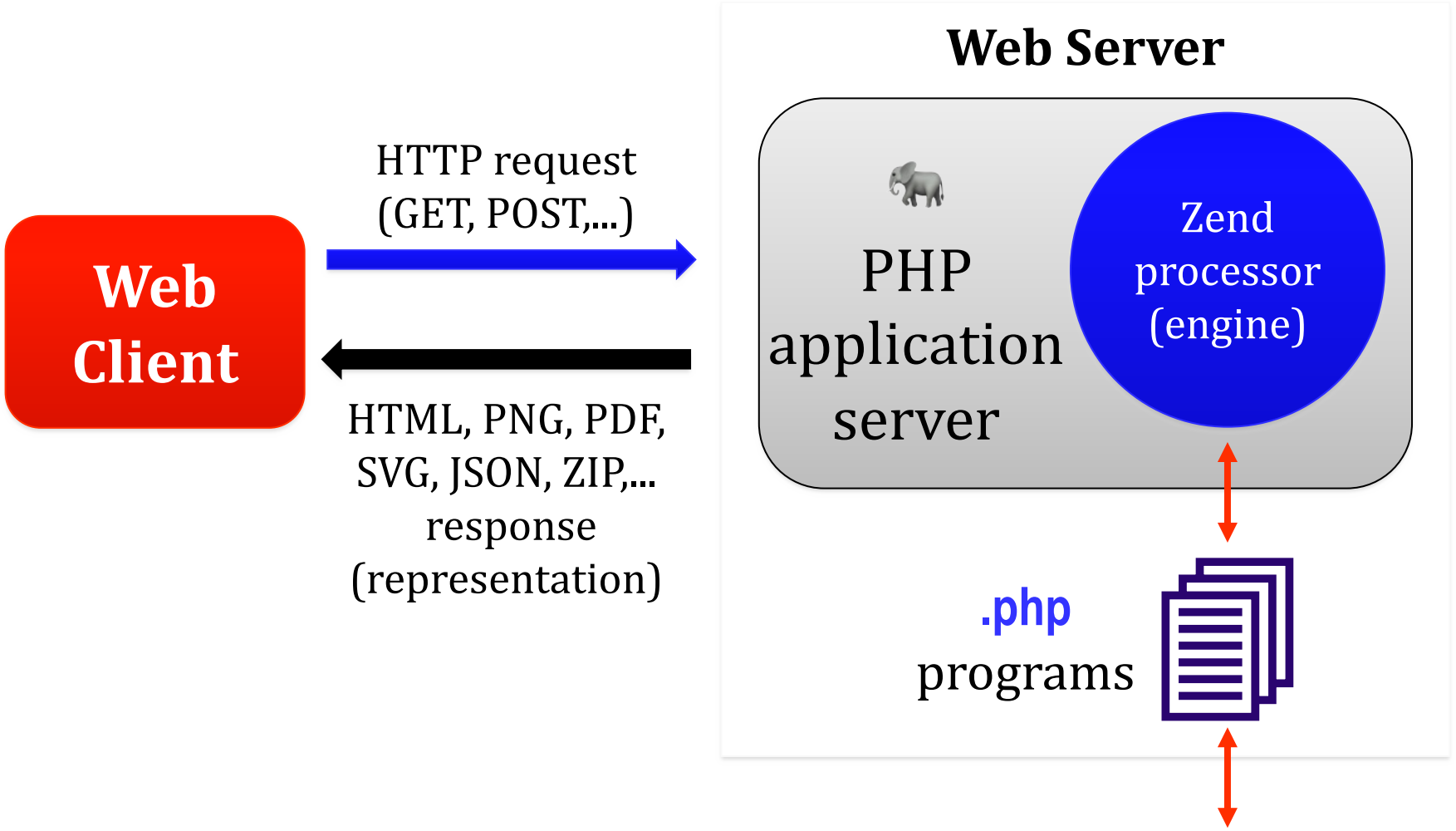
Freely available – open source – for various platforms
(FreeBSD, Linux, Windows, macOS, etc.)
and Web servers: Apache, IIS, Nginx,…

www.php.net
www.zend.com

PHP processor (engine) mode of operation



PHP processor (engine) mode of operation



eventually, various extensions (modules) can be integrated

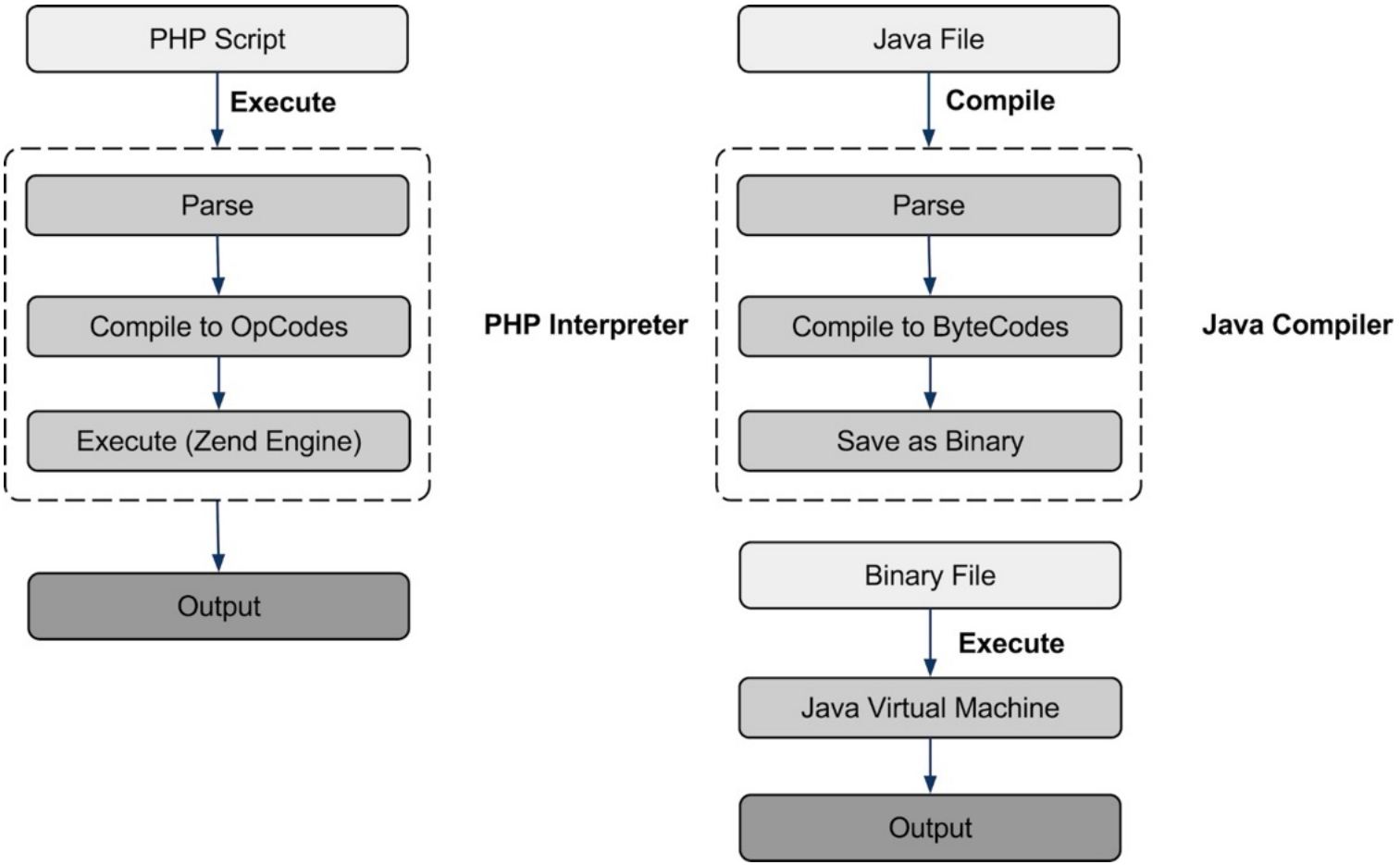
www.php.net/manual/en/extensions.php

php: characteristics

A PHP program is interpreted by Zend Engine 2
which generate internal instructions – opcodes

www.php.net/manual/en/internals2.opcodes.php

J. Pauli, N. Popov, A. Ferrara, *PHP Internals Book*, 2025
www.phpinternalsbook.com



important phases of
interpreting PHP programs *vs.* compiling Java code

PHP code is interpreted each time
when it must be executed

advanced

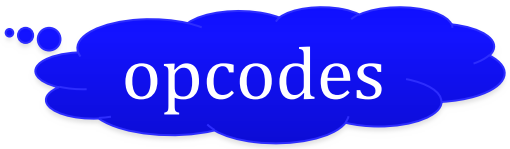
```php
<?php
class Greeting {
    public function sayHello ($to)
    {
        echo "Hello $to";
    }
}

$greeter = new Greeting ();
$greeter->sayHello ("World");
?>
```

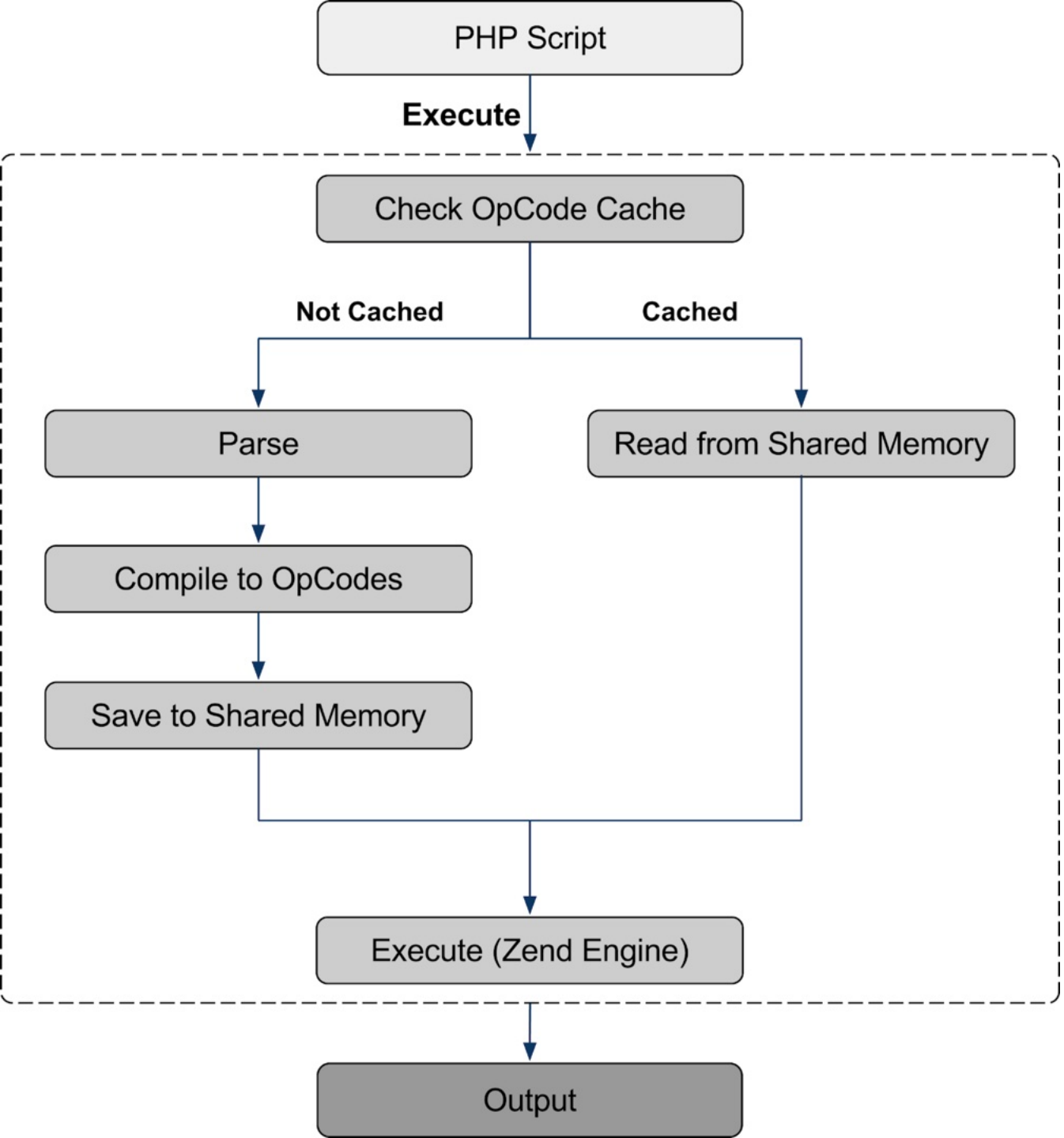
transforming PHP code into opcodes

php.watch/articles/php-dump-opcodes

```
line      # *    op                fetch          ext     return     operands
---------------------------------------------------------------------------
   3      0  >   EXT_NOP
          1       RECV                                        !0
   5      2       EXT_STMT
          3       ADD_STRING                                  ~0     'Hello+'
          4       ADD_VAR                                     ~0     ~0, !0
          5       ECHO                                               ~0
   6      6       EXT_STMT
          7     > RETURN                                             null
```



advanced



for efficiency, the opcodes are stored into a shared memory

```
precision = 14            ; precision for float values – details at php.net/precision
safe_mode = Off           ; processing control – study php.net/safe-mode
disable_functions = "allow_url_fopen, eval" ; unallowed functions (e.g., security reasons)
max_execution_time = 30       ; max. number of seconds allocated to program execution
memory_limit = 128M  ; max. size of the memory allowed for a script
post_max_size = 8M    ; max. size of data transmitted via POST method
default_mimetype = "text/html" ; default MIME type for the content generated by a script
file_uploads = On         ; file uploads are permitted
upload_max_filesize = 32M    ; max. size of an uploaded file
session.use_cookies = 1       ; Web sessions will be stored in cookies
session.name = PHPSESSID ; default cookie name regarding Web sessions
...
; specifying the extensions loaded during the PHP server application initialization
extension_dir="./php/extensions" ; directory including extensions
extension=curl                ; HTTP + other Internet protocols using libcurl library
extension=pdo_sqlite          ; support for SQLite via PDO (PHP Data Objects)
extension=mysqli              ; support for MySQL
...
```

various behaviors of PHP platform,
including loaded extensions (.so/.dll shared libraries),
can be globally configured by using the **php.ini** file

php: characteristics

PHP program execution behavior can be adjusted via **declare** directive – possibly, for a block of code

www.php.net/manual/en/control-structures.declare.php

```
// charset encoding used for generated content
declare (encoding='UTF-8');

// strict datatype checking – PHP 7+
declare (strict_types=1);
```

advanced

php: characteristics

To increase performance, the just-in-time compilation (JIT) could be adopted

HHVM – HipHop Virtual Machine (Facebook)

PHP source-code ▸ opcodes ▸ machine code (e.g., x86-64)

JIT

www.hhvm.com

advanced

php: characteristics

To increase performance, the just-in-time compilation (JIT) could be adopted

HHVM – HipHop Virtual Machine (Facebook)

includes a high-performance server: Proxygen
allows the upload of extensions written in PHP/C++
used by Baidu, Box, Etsy, Facebook, Wikipedia,…
research.facebook.com/publications/the-hiphop-virtual-machine/

advanced

php: characteristics

To increase performance, the just-in-time compilation (JIT) could be adopted

PHP 8 offers native support

www.zend.com/blog/exploring-php-8
www.zend.com/blog/exploring-new-php-jit-compiler

php: characteristics

User interaction:

retrieving the values entered into Web form fields
cookies
sessions
user authentication

access to global variables – created “on the fly”

php: characteristics

Features regarding Web technologies:

URL processing
HTTP support – including cURL
caching via memcached
Web services development – e.g. REST APIs

…and others

php: characteristics

Support for database access:

abstract layers

DBAL (DataBase Abstraction layer)
iODBC (Independent Open DataBase Connectivity)
PDO (PHP Data Objects)

www.phptherightway.com/#databases_abstraction_layers

php: characteristics

Support for database access:

specific to a database server

relational: DB2, MySQL, Oracle, PostgreSQL, SQLite,…
NoSQL – e.g., MongoDB

consult **www.phptherightway.com/#databases**

php: characteristics

Resource content processing:

audio files – via various libraries: ktaglib, oggvorbis, etc.
bzip2, LZF, RAR, ZIP, ZLIB archives
PDF documents
graphical content in various formats
JSON files
XML documents – creating, processing, validating, etc.
information regarding credit cards
...

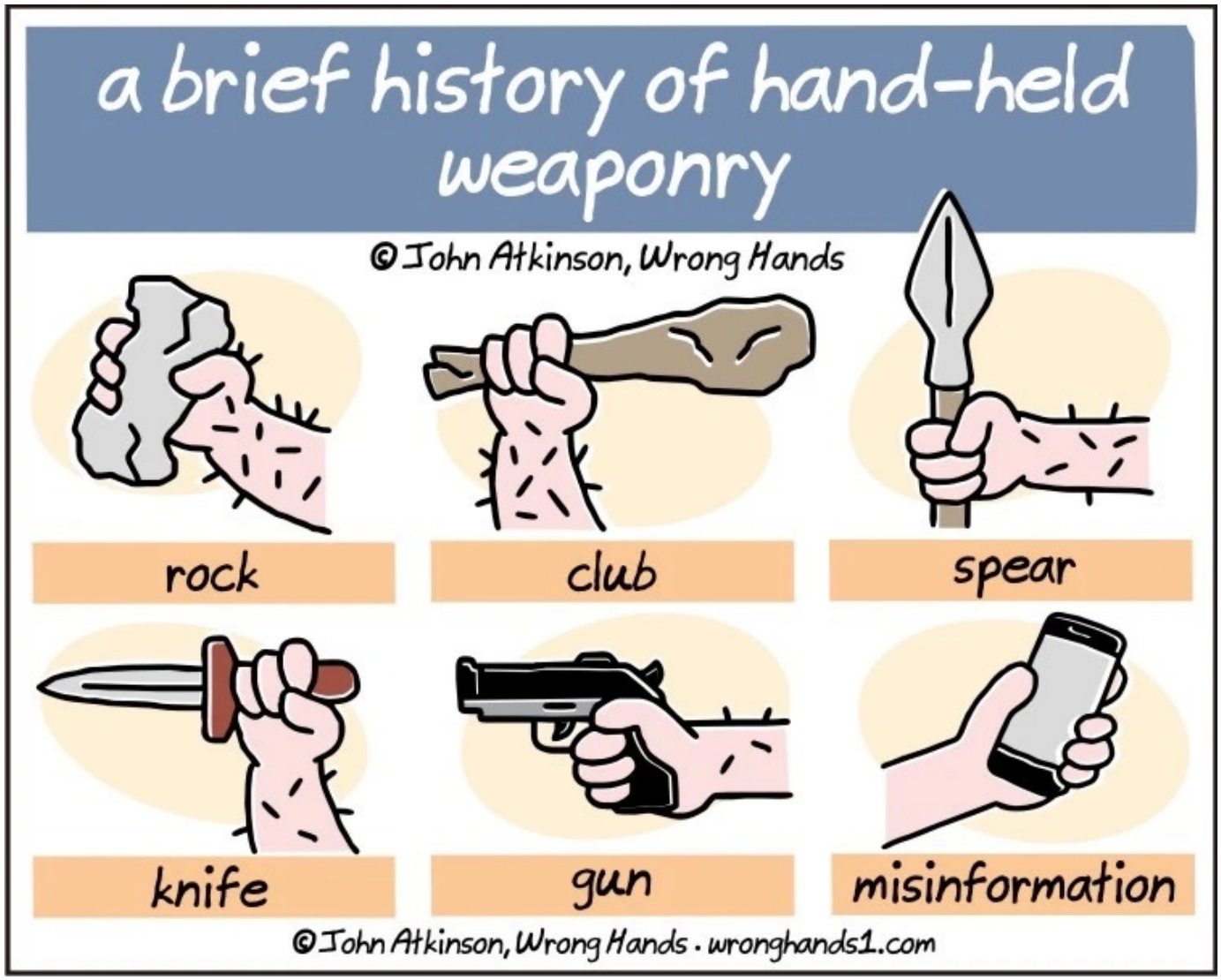
php: characteristics

Support for Internet + system resources:

file systems, including FTP
processes – with Libevent, PCNTL, pthreads,...
event processing – via Event
sockets
e-mail – e.g., IMAP, POP3

...and many others

(instead of) break



wronghands1.files.wordpress.com/2023/01/a-brief-history-of-hand-held-weaponry.jpg

PHP as procedural programming language

php: characteristics

Procedural programming

paradigm based on procedure calls (routines, functions) containing a series of steps for making the calculations

procedural languages are imperative, using instructions (commands) that change the state of the program

examples: FORTRAN (1954), ALGOL (1958), BASIC (1964), Pascal (1970), C (1972), Ada (1978)

php: data types – scalars

boolean

true or false

php: data types – scalars

int

integer values specified in…

10 (decimal)	3734
16 (hex)	0xE96
8 (octal)	07226
2 (binary)	0b111010010110

www.php.net/manual/en/language.types.integer.php

php: data types – scalars

float

real numbers
usually, conforming to IEEE 754 (double precision)

the size of the representation is platform dependent
(the precision can be adjusted in the config file php.ini)

www.php.net/manual/en/language.types.float.php
floating-point-gui.de

php: data types – scalars

float

special value:
NAN (not a number) constant

useful predefined functions:
is_nan()
is_finite()
is_infinite()

php: data types – scalars

string

ASCII character strings
(starting with PHP 7, native Unicode support exists)

www.php.net/manual/en/language.types.string.php

php: data types – scalars

string

ASCII character strings
(starting with PHP 7, native Unicode support exists)

escape characters can be used, like
\n (New Line) **\r** (Carriage Return) **\t** (Tab)
\\ (Backslash) **\$** (Dollar) **\"** (Double Quote) **\'** (Quote)

php: data types – scalars

string

common delimiters:
" or '

php: data types – composed

array

association between values (of any type)
and keys (of integer or string types)

www.php.net/manual/en/language.types.array.php

php: data types – composed

array

there is no clear distinction
between indexed and associative arrays

an array could represent various data structures:
list (vector), associative array - hash (implementation of an association of values - mapping), dictionary, collection, stack, queue,…

php: data types – composed

array

```
// an indexed array (vector of values) – initial syntax
$gifts = array (“watch", “cookie", “necklace", "axe");

// associative array – <key, value> pairs
array ( "name" => "Tux", "size" => 17, "offer" => TRUE );

// simplified syntax (starting with PHP 5.4)
[ "name" => "Tux", "size" => 17, "offer" => TRUE ];
```

php: data types – composed

object

instance of a class

created with the **new** operator

www.php.net/manual/en/language.types.object.php

php: data types – special

resource

denotes a reference to an external resource

examples: bzip2, curl, ftp, gd, mysql link, mysql result, pdf document, printer, stream, socket, xml, zlib

a resource is created by using specific functions
e.g., a stream resource is initiated by the fopen() function and used by functions like fread(), feof(), fgets(), etc.

php: data types – special

resource

denotes a reference to an external resource

predefined functions:
is_resource()
get_resource_type()

details at www.php.net/manual/en/resource.php

php: data types – special

null

specifies the null value
representing a variable that has no value

useful functions:
is_null()
unset()

php: variables

Variables have names composed by letters, digits, and _ characters prefixed by the **$** symbol

can store values – having a specific data type – or references (specified with **&**)

www.php.net/manual/en/language.variables.php

php: variables

Variables are created “on the fly”
data type is inferred according to the context

data type automatic conversion (type casting)
is similar to the C language

```
$age = 21;                      /* a variable of Integer type */
$connected = TRUE;              # a Boolean datatype one
$prefer["color"] = "gray";      // an associative array
```

```php
$nume = 'Tux';

// different behavior regarding the substitution of the actual
// value of a variable depending on the delimiters used
// for strings
echo "Salut $nume!\n";
echo 'Salut $nume!\n';
```

-> Salut Tux!
Salut $nume!\n

</> source code

```php
<?php
$nume = 'Tux';

echo "Salut $nume!\n";
echo 'Salut $nume!\n';
```

input Output

Success #stdin #stdout 0.02s 24312KB

```
Salut Tux!
Salut $nume!\n
```

online execution of PHP code via Ideone Web tool

php: variables

Useful predefined functions:

var_dump()

settype()

is_bool(), is_int(), is_float(), is_array(), is_string()

is_scalar(), is_numeric()

...and others

php: variables

Scope (variables visibility)

in order to be accessed in the entire program,
the variables should be declared as **global**

php.net/manual/en/language.variables.scope.php

```php
$score = 33;

function getScore () {
   echo "Current score: " . $score;
}

getScore();
```

- Undefined variable:
  score in prog.php on line 4

```php
$score = 33;

function getScore () {
   global $score;
   echo "Current score: " . $score;
   // similar to $GLOBALS["score"]
}

getScore();
```

- Current score: 33

php: variables

Scope (variables visibility)

a variable can also be declared as **static**

exists only in a local scope (e.g., inside a function),
but it does not lose its value
when program execution leaves this scope

php: variables

Allocated memory is automatically freed (garbage collection)

each variable has a container associated with it in memory (*zval*)

see xdebug_debug_zval() function provided by Xdebug extension

www.php.net/manual/en/features.gc.php

php: predefined variables

Variables available in the whole program (superglobals)

$GLOBALS [ ]

an associative array containing references to all variables defined as global

php: predefined variables

$_SERVER [ ]

$_GET [ ] $_POST [ ] $_FILES [ ] $_REQUEST [ ]

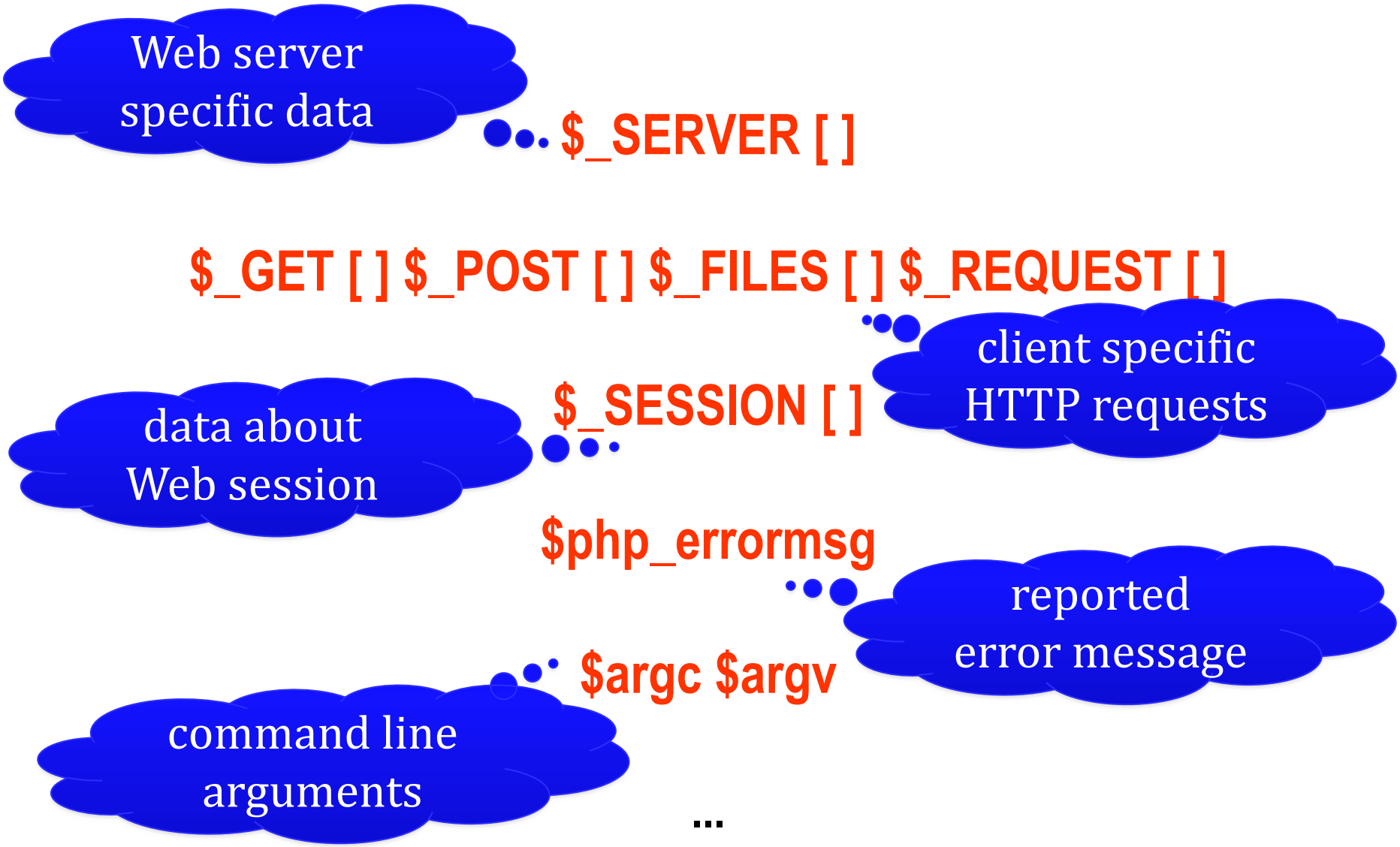
$_SESSION [ ]

$php_errormsg

$argc $argv

...

php: predefined variables



php: constants

Specified with **define ( )**

globally available in the program

define (string $name , mixed $value) : bool

```
define ("MIN_DIMENS", 13);
```

www.php.net/manual/en/function.define.php

php: predefined constants

Examples:

PHP_VERSION
PHP_OS
PHP_EOL
PHP_INT_MAX
PHP_INT_SIZE
DIRECTORY_SEPARATOR
true
false
null

php: predefined constants

Error reporting control:

E_ERROR	fatal errors (script execution is ended)
E_WARNING	warnings
E_PARSE	code processing errors (parsing)
E_NOTICE	notifications at runtime
E_STRICT	suggestions on code improvement
E_DEPRECATED	notifications about deprecated aspects

www.php.net/manual/en/errorfunc.constants.php
www.phptherightway.com/#errors_and_exceptions

advanced

php: predefined constants

Runtime environment offers access to “magical” constants
their values can be used in a given program

__LINE__
__FILE__
__DIR__
__FUNCTION__
__CLASS__
__TRAIT__
__METHOD__
__NAMESPACE__

php: operators

Majority of them are similar to the C language ones

arithmetic: **+ - * / % ++ --**
value assignation: **=** and **=>** (for arrays)
reference: **&**
reference assignation: **=&**
bit-wise: **& | ^ << >>**
comparisons: **== === != <> !== < > <= >= ?: ?? <=>**
error reporting control: **@**
logic: **and or xor ! && ||**
string (concatenation) – same as Perl: **. .=**

php: operators

Starting with PHP 7, new operators can be used:

<=> (spaceship)
comparing expressions (of a scalar type),
and returning -1, 0 or 1

```
echo 15.5 <=> 15.5;      // 0  (equality)
echo 15.5 <=> 16.5;      // -1 (right value is greater)
echo 17.5 <=> 15.5;      // 1  (left value is greater)
```

php: operators

Starting with PHP 7, new operators can be used:

?? (null coalescing)
offers the value of first operand if exists and it's not NULL,
otherwise returns the value of second operand

```
// as username, we'll use the value provided by a Web form
// (retrieved with GET or POST);
// if it doesn't exist, the value is 'tux'
$username = $_GET['user'] ?? $_POST['user'] ?? 'tux';
```

php: control statements

if, switch, while, do, for, break, continue
similar to the C instructions

```
if (!$name) {
  echo ("No name was given…");
} else {
  echo ("Welcome, " . $name . "!\n");
}
```

php: example

```
<?php
  // filling up an array with values from 1 to 10
  for ($index = 1; $index <= 10; $index++) {
    $values[$index] = $index;
  }
  // computing the sum of values
  $sum = 0;
  foreach ($values as $item)
    $sum += $item;
  /* showing the obtained sum on the standard output
     to be sent to the Web client */
  echo ("<p>Sum of first 10 numbers is <strong>" .
          $sum . "</strong>.</p>");
?>
```

Invoking (running) the PHP program
from the command line:

saving source-code into a text file – **values.php**

calling PHP interpreter from the command line interface

```
php values.php
<p>Sum of first 10 numbers is <strong>55</strong>.</p>
```

Invoking (running) the PHP program on the Web server:

placing the source code – having read and execution rights

in the browser, specifying the URL to the program to be invoked via the GET method of the HTTP protocol

result generated by the script

php: control statements

Including source-code from other files (support for modularization)

include

searches the source file in predefined directories specified via include_path and evaluates it

if the file does not exist, a warning is generated

include_once – to include it only once

php: control statements

Including source-code from other files
(support for modularization)

require

searches the source file in predefined directories specified via include_path and evaluates it

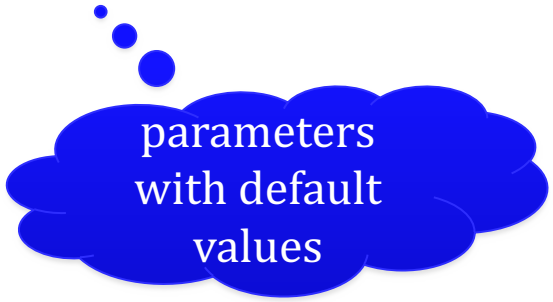
if the file does not exist, a fatal error is generated

require_once – to include it only once

php: functions

User defined functions:

```
function sendMessage ($to="", $from="", $subject="Web") {
   // body…
}
```



php.net/manual/en/language.functions.php

```php
define ('MAX', 10);                    // maximum number of values

function square ($number) {  // a function to square a number
  return $number * $number;
}

$number = 0;
while ($number < MAX) {
  $number++;                           // incrementing…

  if ($number % 2)                     // is odd number...
    continue;                          // continuing to next iteration
  // it is an even number, so we show its square
  echo "Square $number is " . square ($numar) . "\n";
} // end of while
```

php: functions

User defined functions:

a function name is case-insensitive

parameters can be specified by reference – prefixed by &

a variable number of parameters is indicated by ...

php.net/manual/en/functions.arguments.php

⚠ The functions are visible anywhere in the program:

```
infomagic ();

function infomagic () {
  printf("<p>I'm <code>%s()</code> and I'm executing line %d
    from <code>%s</code>.</p>", __FUNCTION__, __LINE__, __FILE__);
  printf("<Using PHP <code>%s</code> on <code>%s</code>
   with the Web server <code>%s</code>.</p>",
   PHP_VERSION, php_uname(), $_SERVER['SERVER_SOFTWARE']);
}
```

```php
<?php
infomagic ();

function infomagic () {
  printf("<p>Sunt <code>%s()</code> și execut linia %d din <code>%s</code>.</p>",
    __FUNCTION__, __LINE__, __FILE__);
  printf("<p>Folosesc PHP <code>%s</code> de pe <code>%s</code>.</p>",
    PHP_VERSION, php_uname());
}
```

eval(); 8.5.3

shortcut: ctrl+enter

PHP Sandbox

Execute

took 44 ms, 16.41 MiB

```
linia 6 din <code>/in/BJbk1</code>.</p>
code>Linux php_shell 4.8.6-1-ARCH #1 SMP PREEMPT Mon
```

PHP Versions and Options (8.4.10)

Other Options

Execute Code | Save or share code

Result for 8.4.10: Execution time: 0.000134s Mem: 3

```
<p>Sunt <code>infomagic()</code> și execut linia 7
  din <code>/home/user/scripts/code.php</code>.</p><p>Folosesc PHP
<code>8.4.10</code> de pe <code>Linux ip-172-31-20-47.eu-
central-1.compute.internal 5.10.214-202.855.amzn2.x86_64 #1 SMP Tue
Apr 9 06:57:12 UTC 2024 x86_64</code>.</p>
```

rapid testing of the code:

PHP Sandbox

onlinephp.io

PHPTester

phptester.net

```php
<?php
declare (strict_types=1);

// arguments should be integers, returned value must be of integer type
function add (int ...$numbers): int {
    $sum = 0;
    foreach ($numbers as $number) {
        $sum += $number;
    }
    return $sum;
}

echo add (7, 3, 74, 1);
echo add (pi (), '?');
?>
```

85

Fatal error: Uncaught TypeError: Argument 1 passed to add() must be of the type integer, float given
Next TypeError: Argument 2 passed to add() must be of the type integer, string given

PHP 7+: data type can be specified for each argument and for the value returned by the function (scalar type declarations)

php: functions

Anonymous functions can also be defined

- functional programming (e.g., closures)

```
$hello = function ($name) {          // variable of type function
    printf ("Hello %s...\n", $name);  // without collateral effects
};

$hello ('world');
$hello ('Tuxy');
```

www.phptherightway.com/pages/Functional-Programming.html

php: functions

Anonymous functions
can be specified in a shorter way
starting with PHP 7.4

- **short closures**

concept similar to arrow functions

functional construction available
in the JavaScript language (ECMAScript 6 – ES6)

```php
declare (strict_types=1);                // strict data type check
header ('Content-type: text/plain'); // send unformatted text to the client

// short declaration of an anonymous function
// which has one string argument and returns an int
// variable of function type: fn(argument) => body
// (for the body, a single expression is allowed; 'return' is not allowed)
$hello = fn(string $name): int => printf("Hello %s! [la %s].\n",
                                          $nume, date("j-m-Y H:i:s T", time()));
$hello('world');
sleep (2); // wait 2 seconds...
$hello('Tuxy');
```

```
Hello world! [at 20-03-2025 04:24:31 UTC].
Hello Tuxy! [at 20-03-2025 04:04:33 UTC].
```

php: predefined functions

mathematical & conversion
character string manipulation
array processing
access to resources and file processing
database manipulation
regarding network connections
cryptographic
XML, PDF, JPEG,... resource processing
operating system specific
general purpose

details at **php.net/manual/en/funcref.php**

php: predefined functions

Math:

abs(), mod(), fmod()
ceil(), floor(), round(), max(), min()
exp(), log10(), log()
pow(), sqrt()
sin(), cos(), tan(), asin(), …, sinh(), …, pi()
rand(), srand()
bindec(), octdec(), dechex(),…, base_convert()
is_finite(), is_infinite(), is_nan()

php.net/manual/en/refs.math.php

php: predefined functions

Strings:

echo(), print(), printf(), sprintf(), etc.
strlen(), chr(), ord(), substr(), strstr(), strpos(),…
strcmp(), strcasecmp(), strnatcmp(), etc.
strcat(), str_replace(), str_ireplace(), strrev(), etc.
trim(), ltrim(), rtrim()
explode(), implode(), split(), join(), strtok()

PHP 8: str_contains(), str_starts_with(), str_ends_with()

details about text processing:
php.net/manual/en/refs.basic.text.php

php: predefined functions

Regular expressions:

conforming to POSIX standard
ereg(), ereg_replace(), split(), etc.

Perl compatible – PCRE: **www.pcre.org**
preg_filter(), preg_grep(), preg_match(), preg_split(),…

php: predefined functions

Regular expressions:

conforming to POSIX standard
ereg(), ereg_replace(), split(), etc.

Perl compatible – PCRE: **www.pcre.org**
preg_filter(), preg_grep(), preg_match(), preg_split(),…

PHP 8+: new **match()** construction
www.php.net/manual/en/control-structures.match.php

php: predefined functions

Arrays:

array_count_values(), array_search(), array_filter(),
array_slice(), array_chunk()
array_fill(), array_combine(), array_shift(),
array_reverse(), array_multisort(), array_sum(),…
array_merge(), array_intersect(), array_diff()
array_keys(), array_key_exists()
array_push(), array_pop()
array_map(), array_reduce()

php.net/manual/en/book.array.php

advanced

```php
/* filters certain values from an array
   by using a programmer defined function */
function value_less_than ($number) {
   // returns an expression of function type
   return function ($item) use ($number) { // a closure (functional approach)
      return $item < $number;
   };
}

$marks =  [ 7, 8, 9, 10, 7.5, 3, 10, 8.75, 4 ];
// using array_filter(), a predefined function, on marks array
// to obtain the values less than a given value (here: 7)
$values = array_filter ($marks, value_less_than (7));

print_r ($values);
```

```
Array
(
    [5] => 3
    [8] => 4
)
```

also study wiki.php.net/rfc/closures

advanced

In PHP 7.4+ the operator ... can be used within arrays (similar with ECMAScript 2018)

```php
$web = [ 'JS', 'PHP' ];
$whims = [ 'Lua', 'Rust', 'Scala' ];
$languages = [ 'C#', ...$web, 'Java' ];

print_r ($languages);
// fusion – similar with array_merge()
print_r ([ ...$whims, ...$web ]);
```

```
Array
(
    [0] => C#
    [1] => JS
    [2] => PHP
    [3] => Java
)
Array
(
    [0] => Lua
    [1] => Rust
    [2] => Scala
    [3] => JS
    [4] => PHP
)
```

php: predefined functions

Arrays

new functions available since PHP 8.4:

array_find()

array_find_key()

array_any()

array_all()

wiki.php.net/rfc/array_find

new functions in PHP 8.5 – 2025:

array_first()

array_last()

wiki.php.net/rfc/array_first_last

php: predefined functions

Character manipulation:

ctype_digit(), ctype_xdigit(), ctype_print(), ctype_punct(), ctype_space(),…

ctype_alpha(), ctype_alnum(), ctype_lower(), ctype_upper()

php: predefined functions

Date & time:

getdate(), localtime(), gettimeofday(), time(), etc.
date(), idate(), gmdate(),…
checkdate()
strftime(), strtotime()

see also Calendar, DateTime, HRTime extensions

php: predefined functions

PHP variables:

empty(), isset(), unset()
strval(), print_r(), var_dump()
serialize(), unserialize()

also, consult php.net/manual/en/book.var.php

php: predefined functions

Files and directories:

using FILE data type – similar to C language:
fopen(), fread(), fscanf(), fgets(), fwrite(), fprintf(),
fseek(), ftell(), feof(), fclose(), ftruncate(), fstat(),...

file(), copy(), rename(), delete(),
move_uploaded_file(), tmpfile()

file_exists(), filesize(), filetype(), fileperms(),…, stat()

is_dir(), is_file(), is_readable(), is_writeable(),…

chdir(), mkdir(), rmdir()

disk_free_space(), disk_total_space()

study **php.net/manual/en/refs.fileprocess.file.php**

php: predefined functions

URLs:

urldecode(), urlencode(), parse_url()
base64_decode(), base64_encode()
http_build_query()

starting with PHP 8.5,
the URL manipulation can be done via the URI extension
thephp.foundation/blog/2025/10/10/php-85-uri-extension/

processing the components of a URL:

```
define('URL', 'http://somewhere.info:8080/offer/toys/product/?name=Tux&size=17#offer');

header('Content-type: text/plain'); // content sent to the client will be plain text

// dividing the URL into components of an associative array
$WebAddress = parse_url(URL);

foreach(array_keys($WebAddress) as $component) {
  printf("%s=%s\n", $component, $WebAddress[$component]);
}

// processing the URL's query string
// each parameter will be stored in $parameters array
parse_str($WebAddress['query'], $parameters);

if ($parameters['size'] < 13) {
  echo "Toy is not ok...\n";
} else {
  echo "Toy is ok (" . $parameters['size'] . ")
          and has the name " . $parameters['name'] . ".\n";
}
```

processing the components of a URL:

```
define('URL', 'http://somewhere.info:8080/offer/toys/product/?name=Tux&size=17#offer');

header('Content-type: text/plain'); // content sent to the client will be plain text

// dividing the URL into components of an associative array
$WebAddress = parse_url(URL);

foreach(array_keys($WebAddress) as $component) {
  printf("%s=%s\n", $component, $WebAddress[$component]);
}

// processing the URL's query string
// each parameter will be stored in $parameters array
parse_str($WebAddress['query'], $parameters);

if ($parameters['size'] < 13) {
  echo "Toy is not ok...\n";
} else {
  echo "Toy is ok (" . $parameters['size'] . ")
          and has the name " . $parameters['name'] . ".\n";
}
```

```
scheme=http
host=somewhere.info
port=8080
path=/offer/toys/product/
query=name=Tux&size=17
fragment=offer
Toy is ok (17)
and has the name Tux.
```

php: predefined functions

Web resource processing (HTML, JSON):

nl2br(), htmlentities(), htmlspecialchars(), strip_tags()

get_browser(), show_source(), highlight_string(),...

json_encode(), json_decode(), json_last_error()

(de/en)coding internal data ↔ JSON data:

```
// songs from the album 'Animals' by Pink Floyd (1977) expressed in Unicode
$animals = [ "🐷 on the Wing #1", "Dogs 🐕",
  [ "🐖🐖🐖" => "Pigs (Three Different Ones)" ],
  "Sheep 🐑", "🐷 on the Wing #2" ];

$json = json_encode($animals);   // encoding (serializing) in JSON

$animals2 = json_decode($json);  // decoding JSON data

// information regarding used variables
var_dump($animals, $animals2, $json);
```

$animals

```
array(5) {
  [0]=>
    string(19) "🐷 on the Wing #1"
  [1]=>
    string(16) "Dogs 🐕"
  [2]=>
    array(1) {
    ["🐖🐖🐖"]=>
      string(27) "Pigs (Three Different Ones)"
  }
  [3]=>
    string(10) "Sheep 🐑"
  [4]=>
    string(19) "🐷 on the Wing #2"
}
```

$animals2

```
array(5) {
  [0]=>
    string(19) "🐷 on the Wing #1"
  [1]=>
    string(16) "Dogs 🐕"
  [2]=>
    object(stdClass)#1 (1) {
    ["🐖🐖🐖"]=>
      string(27) "Pigs (Three Different Ones)"
  }
  [3]=>
    string(10) "Sheep 🐑"
  [4]=>
    string(19) "🐷 on the Wing #2"
}
```

$json

string(191) "["\ud83d\udc37 on the Wing #1","Dogs \ud83d\udc15\u200d\ud83e\uddba",
{"\ud83d\udc16\ud83d\udc16\ud83d\udc16":"Pigs (Three Different Ones)"},
"Sheep \ud83d\udc11","\ud83d\udc37 on the Wing #2"]"

php: predefined functions

Support for cryptographic operations:

password hashing – password_*() functions

useful extensions

- Hash (hash_*() digest functions)
- OpenSSL (functionalities for SSL/TLS)
- Sodium (advanced encrypt/decrypt operations)
- Random (work with pseudo-random numbers – in PHP 8.2+)

php.net/manual/en/refs.crypto.php

php: predefined functions

Support for graphical content (raster/vector):

preinstalled extensions

Cairo – vector/raster processing: www.cairographics.org

EXIF – access to JPEG meta-data

GD – raster processing (GIF, JPEG, PNG): libgd.github.io

ImageMagick – multi-format processing: www.imagemagick.org

www.php.net/manual/en/refs.utilspec.image.php

php: predefined functions

Other useful functions:

die(), eval(), exit(), sleep(), usleep(), time_sleep_until()

uniqid(), sys_getloadavg()

php_info(), php_check_syntax()

advanced

php: other features

SPL (Standard PHP Library)
access to standard ways of data processing

defined data structures:
SplStack, SplQueue, SplHeap, SplPriorityQueue,...

iterators:
ArrayIterator, FilesystemIterator, RegexIterator, etc.

www.php.net/spl
www.phptherightway.com/#standard_php_library

advanced

php: other features

Internationalisation & localization of applications

lexical checking (spelling): Enchant

support for translating messages: Gettext

internationalisation (i18n): intl
classes: NumberFormatter MessageFormatter IntlDateFormatter IntlTimeZone IntlCalendar Locale

www.php.net/manual/en/refs.international.php

advanced

php: other features

Other preinstalled basic extensions

support for efficient data structures

interfaces:
Collection Hashable Sequence

classes:
Vector Deque Map Pair Set Stack Queue PriorityQueue

www.php.net/manual/en/book.ds.php

advanced

php: other features

Other preinstalled basic extensions

DS – various data structures: Map, Pair, Set, Stack, Vector,...
FANN (*Fast Artificial Neural Network*) – neural networks
GeoIP – geographical localization based on IP address
SeasLog – journaling features (logging)
Streams – work with data streams
Swoole – network access: TCP, UDP, HTTP, WebSocket
YAML – support for *YAML Ain't Markup Language*

www.php.net/manual/en/refs.basic.other.php

advanced

php: other features

Process execution control

extensions of interest
(some available only on Linux systems)

Eio, Ev, Expect, Libevent, PCNTL, POSIX, parallel, pthreads, pht, Semaphore, Shared Memory, Sync

www.php.net/manual/en/refs.fileprocess.process.php

advanced

php: other features

Process execution control

Asynchronous code execution can be done via additional libraries

example:
ReactPHP
facilitates asynchronous – non-blocking – event-based input/output operations
reactphp.org

advanced

php: other features

Program execution from command line

PHP CLI

www.php.net/manual/en/features.commandline.php

advanced

php: other features

Embedded Web server

run with:

php –S localhost:8000 –t *phpwebapp/*

www.php.net/features.commandline.webserver

How about PHP support
for object-oriented programming?

php: characteristics

Object-oriented programming

paradigm based on the concept of **object** – including **data** (attributes, properties) and **code** (methods, procedures)

usually, objects interact with each other and represent **class** instances

examples: Smalltalk (1972), Objective-C (1984), C++ (1985), Python (1990), Java (1995), C# (2000)

php: classes

Class definition via **class**
and instantiation by using **new** operator

objects are treated similar to references
(a variable of object type contains a reference to an object, not a copy of it)

www.php.net/language.oop5

object-oriented programming – encapsulation

```
class Student { // specifying a class
  // properties (data members)
  private $year;
  private $email;
  public  $name;
  // public methods
  public function setYear ($aYear) {
     $this->year = $aYear;
  }
  public function getYear () {
     return $this->year;
  }
}
```

$this is a pseudo-variable
specifying a reference to the current object

object-oriented programming – encapsulation

```
class Student { // specifying a class
  // properties (data members)
  private $year;
  private $email;
  public  $name;
  // public methods
  public function setYear ($aYear) {
     $this->year = $aYear;
  }
  public function getYear () {
     return $this->year;
  }
}
```

```
// an object instance
$stud = new Student ();
$stud->setYear (2);
$stud->name = 'Tux';
print_r ($stud);
```

```
▸ Student Object
(
    [year:Student:private] => 2
    [name] => Tux
    [email:Student:private] =>
)
```

php: classes

Similar to C++, members – properties or methods – can be declared as

public
private
protected

object-oriented programming – inheritance

```
class CleverStudent extends Student {
    private $marks;    // obtained marks (property)

    public function setMarks ($m) {
        $this->marks = (array) $m;
    }
    public function getMarks () {
        return (array) $this->marks;
    }
}

$otherStud = new CleverStudent ();
// calling a method from base class
$otherStud->setYear (2);
// calling a method from derived class
$otherStud->setMarks (
   ['Web' => 10, 'SoftEng' => 9]
);
```

object-oriented programming – inheritance

```
class CleverStudent extends Student {
    private $marks;    // obtained marks (property)

    public function setMarks ($m) {
        $this->marks = (array) $m;
    }
    public function getMarks () {
        return (array) $this->marks;
    }
}

$otherStud = new CleverStudent ();
// calling a method from base class
$otherStud->setYear (2);
// calling a method from derived class
$otherStud->setMarks (
   ['Web' => 10, 'SoftEng' => 9]
);
```

```
print_r ($altStud);

 ▸ CleverStudent Object
(
    [marks:CleverStudent:private]
        => Array
        (
            [Web] => 10
            [SoftEng] => 9
        )

    [year:Student:private] => 2
    [name] =>
    [email:Student:private] =>
)
```

php: classes

Special methods:

constructors are named __construct()

destructors are named __destruct()

php: classes

Accessing static, constant, or overloaded properties/methods

::

scope resolution operator (Paamayim Nekudotayim)

www.php.net/manual/en/language.oop5.paamayim-nekudotayim.php

php: classes

Accessing static, constant, or overloaded properties/methods

::

self – current class
parent – parent class

starting with PHP 7.4 the data type of the properties defined by a class can be explicitly specified

```php
declare (strict_types=1);

class Student {
  // properties having a type
  private int $year;
  private string $email;
  public  string $name;
  // etc.
}

$stud = new Student ();
$stud->name = true;
```

Fatal error: Uncaught TypeError: Cannot assign bool to property Student::$name of type string

php: classes

Properties or methods declared as **static**
can be accessed without class instantiation

 for examples, study
www.php.net/manual/en/language.oop5.static.php

php: classes

Abstract classes/methods are declared with **abstract**

abstract classes cannot be instantiated

each class having at least one abstract method
is considered abstract

abstract methods must be implemented in a child class
(specified by **extends**) of an abstract class

php: interfaces

Specifying methods that will be implemented by a class (similar to Java)

```
// interface regarding a content template
interface iTemplate {
    // setting a variable to be substituted with its value
    public function setVar ($name, $var);
    // getting the template representation
    public function getRep ($template);
}
```

www.php.net/manual/en/language.oop5.interfaces.php

```php
// a class implementing the interface
class Template implements iTemplate {
    // associative array including variables to be replaced by their values
    private $variables = array ();

    public function setVar ($name, $var) {
        $this->variables[$name] = $var;
    }

    public function getRep ($template) {
        foreach ($this->variables as $name => $val) {
            // replacing a variable name with its value
            $template = str_replace ('{' . $name . '}', $val, $template);
        }
        return $template;
    }
}
```

advanced aspects: www.phptherightway.com/#templating

php: predefined interfaces & classes

Traversable
Iterator
IteratorAggregate
Throwable
ArrayAccess
Serializable
Closure
Generator

www.php.net/manual/en/reserved.interfaces.php

php: predefined interfaces & classes

```
// example: Iterator interface
Iterator extends Traversable {

  // methods to be written by programmer
  // in the class implementing the interface
  abstract public mixed current ( void )
  abstract public scalar key ( void )
  abstract public void    next ( void )
  abstract public void    rewind ( void )
  abstract public boolean valid ( void )
}
```

php: reflection

Access to information regarding a class:

a PHP program can get data about classes, interfaces, functions, methods, extensions – *reverse engineering*

ReflectionClass implements Reflector

www.php.net/manual/en/book.reflection.php

php: reflection

```
$a_class = new ReflectionClass (‘CleverStudent');
// display information about the specified class
printf ("<p>Class <em>%s</em> extends %s
        and is declared in file <tt>%s</tt>.</p>",
        $a_class->getName (),
        var_export ($a_class->getParentClass (), 1),
        $a_class->getFileName ());
```

Class *CleverStudent* extends
ReflectionClass::__set_state(array('name' => '**Student**',))
and is declared in file **/home/profs/busaco/html/php/introspect.php**

php: traits

Trait

concept borrowed from Self language

collection of methods that can be reused in other classes

www.php.net/manual/en/language.oop5.traits.php

php: traits

Trait

considered as a (C++) class template

in contrast to interfaces, offers implementations of methods, not just their signature

thus, support is provided for multiple pseudo-inheritance

formal aspects: scg.unibe.ch/research/traits

```
// traits (behaviors) that will be associated to 2D geometrical
    shapes
trait Rotating {
  public function rotate ($angle) {      // implements rotation
  }
}

trait Moving {
  public function moveTo ($x, $y) {    // move to other coordinates
  }
}

trait Filling {
  public function fill ($color) {          // performs filling
  }
}
```

```php
abstract class Shape {  // geometrical shapes class
  public function draw() {
    echo ('I am drawing a ' . get_class());
  }
}

class Rectangle extends Shape {  // uses desired traits
  use Filling, Moving, Rotating;      // can be filled, moved, rotated

  public function transform () {    // plus, a specific transformation
  }
}
```

```
// Circle class cannot be extended
final class Circle extends Shape {
  // a circle can be moved or filled
  use Moving, Filling;

  // constant declaration
  const PI = 3.1415265;

  // method specification
  public function computeArea () {
  }
}
```

php: traits

```
// 2 shapes (a circle and a rectangle) are instantiated
$aCircle = new Circle ();
$aRect = new Rectangle ();
$aCircle->draw ();
$aCircle->rotate ();                    // an error will be emitted
$aRect->draw ();
```

- I am drawing a Circle

  PHP Fatal error: Call to undefined method Circle::rotate()
  in /home/dMdWgn/prog.php on line 47

php: special properties

A class has special ("magical") properties associated that can be overloaded

__construct ()
__destruct ()

__toString ()

__get ()
__set ()

others at www.php.net/manual/en/language.oop5.magic.php

php: objects

Objects can be “cloned” via **clone**

Objects can be compared by using the **===** operator

php: objects

Functions for class & object manipulation

get_class() will return an object name,
instance of a given class
get_parent_class() gets the parent class of an object
method_exists() tests if a method exists
for a certain object
class_exists() tests the existence of a class
is_subclass_of() determines if a heritage relation
between two classes exists

php: exceptions

try catch throw

Exception class

similar to Java exceptions

www.php.net/manual/en/language.exceptions.php

php: namespaces

Used to avoid name collisions and to create aliases

declared with **namespace** (first line of a program)

php: namespaces

Used to avoid name collisions and to create aliases

declared with **namespace** (first line of a program)

example: **namespace Facebook;** // Facebook SDK for PHP

www.php.net/language.namespaces

php: namespaces

Used to avoid name collisions and to create aliases

same namespace can be defined in multiple files

namespace hierarchies are permitted

```
namespace Project\Module\Submodule;

class SVGGen { ... };
```

refer with
Project\Module\Submodule\SVGGen

php: namespaces

Used to avoid name collisions and to create aliases

using a specific construct: **use**
(optionally, specifying an alias)

```
use Project\Module\Submodule;
```

php: namespaces

Used to avoid name collisions and to create aliases

using a specific construct: **use**
(optionally, specifying an alias)

```
use Project\Module\Submodule;
```

real examples:

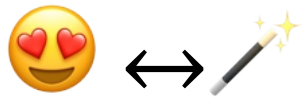
```
use Facebook\Authentication\AccessToken;
use Illuminate\Foundation\Exceptions\Handler as ExceptionHandler;
use Wikimedia\Timestamp\TimestampFormat as TS;
```

php: namespaces

Used to avoid name collisions and to create aliases

without any namespace definition, all class and function definitions are placed into the global space

```
namespace WebProject;
function cosh () {        // specifying our own function
    ...
    $value = \cosh (...); // calling a predefined function (from global space)
}
```



How about the features regarding Web interaction?

php: web interaction

Data transmitted by the client (browser) is stored into (global) predefined associative arrays:

$_GET[ ] – data transmitted via GET

$_POST[ ] – data transmitted via POST

$_COOKIE[ ] – received cookies

$_REQUEST[ ] – data retrieved from client (content of **$_GET**, **$_POST** and **$_COOKIE**)

$_SESSION[ ] – session data

php: web interaction

Other useful global variables:

$_SERVER[ ]
offers info about Web server

$_SERVER['PHP_SELF'] indicates the PHP script name
$_SERVER['REQUEST_METHOD'] – used HTTP method
$_SERVER['HTTP_REFERER'] – URL referring the resource
$_SERVER['HTTP_USER_AGENT'] – client details

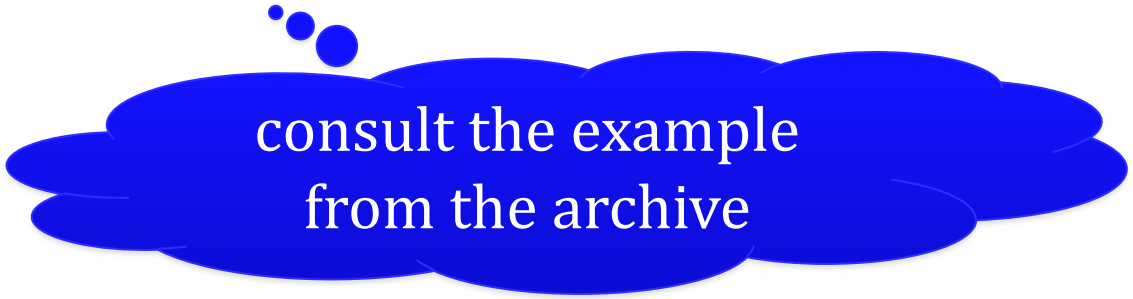
www.php.net/manual/en/reserved.variables.server.php

php: web interaction

Other useful global variables:

$_ENV[ ] – data provided by the runtime environment

$_FILES[ ] – data about the uploaded files
www.php.net/manual/en/features.file-upload.php



```
<!-- a Web form modeled in HTML -->
<form action="display.php" method="post">
        <input type="text" name="name" />
        <input type="text" name="age" />
        <input type="submit" value="Send" />
</form>
```

```
<!-- a Web form modeled in HTML -->
<form action="display.php" method="post">
      <input type="text" name="name" />
      <input type="text" name="age" />
      <input type="submit" value="Send" />
</form>
```

```
<?php
    // display.php – a program invoked via POST
    if (!$_REQUEST["name"])
      display ("Name is not specified!", "err");
    else
      display ("Name is" . $_REQUEST["name"]);
?>
```

each name field in the form represents a key in **$_REQUEST [ ]** associative array (depending on the HTTP method, it could be consulted by using **$_GET[ ]** or **$_POST[ ]**)

php: cookies – creation

 Setting a cookie via **setcookie ()** function

```
setcookie (
   string $name,
   string $value = "",
   int $expires_or_options = 0,
   string $path = "",
   string $domain = "",
   bool $secure = false,
   bool $httponly = false
) : bool
```

Starting with PHP 7.3+, an array of associated options (attributes) can be provided:

```
setcookie (
  string $name,
  string $value = "",
  array $options = [ ]
): bool
```

php: cookies – creation

 Setting a cookie via **setcookie ()** function

```
// persistent – set that the cookie will expire in 10 days
setcookie (‘color', 'tan', time() + 60 * 60 * 24 * 10);

// non-persistent
setcookie ('authenticated', 'true');
```

why?

php: cookies – expiration

 Deleting a cookie:

cancel value and time;

eventually, also other cookie attributes

```
setcookie ($cookie_name, "", 0);
```

www.php.net/manual/en/function.setcookie.php

php: cookies – access

- A cookie is specified (accessed) like a variable of type associative array – $_COOKIE ['cookie_name']

```
<!-- the background color is given by the previously set
     cookie value -->
<style>
   body {
      background: <?php echo $_COOKIE['color']; ?>
   }
</style>
```

see the archive with PHP examples

php: web sessions

- Session management: session_start(), session_register(), session_id(), session_unset(), session_destroy() functions

```
session_start (); // start a Web session
if (!isset ($_SESSION['visits'])) {
        $_SESSION['visits'] = 0; }
else {
        $_SESSION['visits']++; }
```

the variable **visits** attached to the session

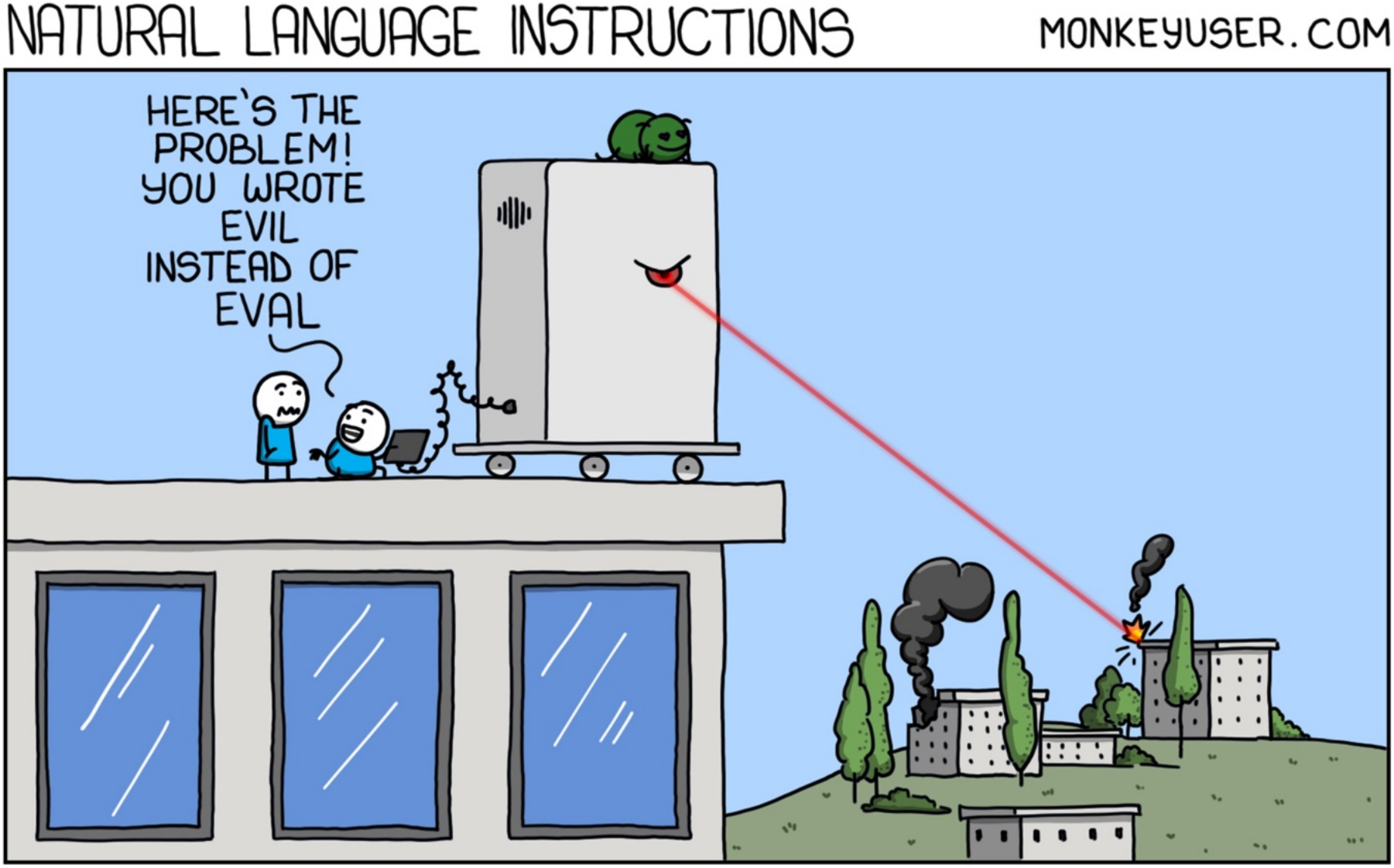
php: web sessions

👽 Session management: SessionHandler class

```
SessionHandler implements SessionHandlerInterface {
  public bool   open ( string $save_path , string $session_name ) : bool
  public string create_sid ( void ) : string
  public string read ( string $session_id ) : string
  public bool   write ( string $session_id , string $session_data ) : bool
  public bool   gc ( int $maxlifetime ) : bool          // remove old sessions
  public bool   destroy ( string $session_id ) : bool
  public bool   close ( void ) : bool
}
```

www.php.net/manual/en/book.session.php

(instead of) break



How can databases be accessed from PHP?

php: db

Native support for multiple database servers/technologies:

MongoDB – classes: **MongoDB MongoClient MongoCursor**
MySQL / MariaDB – **mysqli** class
Oracle –**oci_*()** functions + **OCILob** class
PostgreSQL – **pg_*()** functions + **PgSql** class
SQLite – **SQLite3** class

connections can be persistent

php: db – mysql

Functions/methods to access MySQL/MariaDB

connecting to server: **mysql_connect ()**, **mysql_pconnect ()**

selecting (using) a database: **mysql_select_db ()**

executing a SQL statement: **mysql_query ()**

error reporting: **mysql_errno ()**, **mysql_error ()**

getting results into an array: **mysql_fetch_array ()**

many others…

in the present, **a deprecated approach** – eliminated in PHP 7+

php: db – mysqli extension

Aim: easy and flexible access to MySQL/MariaDB

procedural or object-oriented approach
-> facilitates code maintenance

compatible with MySQL API

ensures security and performance

docs available at **www.php.net/mysqli**

php: db – mysqli extension

Important methods:

initiating a connection to a MySQL server – **mysqli ( )**
issuing SQL statements – **query ( )**, **prepare ( )**, **execute ( )**
processing the response – **fetch ( )**, **fetch_assoc ( )**
closing connection – **close ( )**
etc.

php: db – example

Initially, we'll create a MariaDB account to allow authenticated access from PHP programs to the students database:

```
(console)$ mysql -u root -p mysql
Enter password: **********
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Server version: 10.11.2-MariaDB Source distribution

MariaDB [mysql]> GRANT SELECT, INSERT, UPDATE, DELETE, CREATE,
DROP ON students.* TO 'tux@localhost' IDENTIFIED BY 'p@rola'
WITH GRANT OPTION;
Query OK, 0 rows affected (0.001 sec)
```

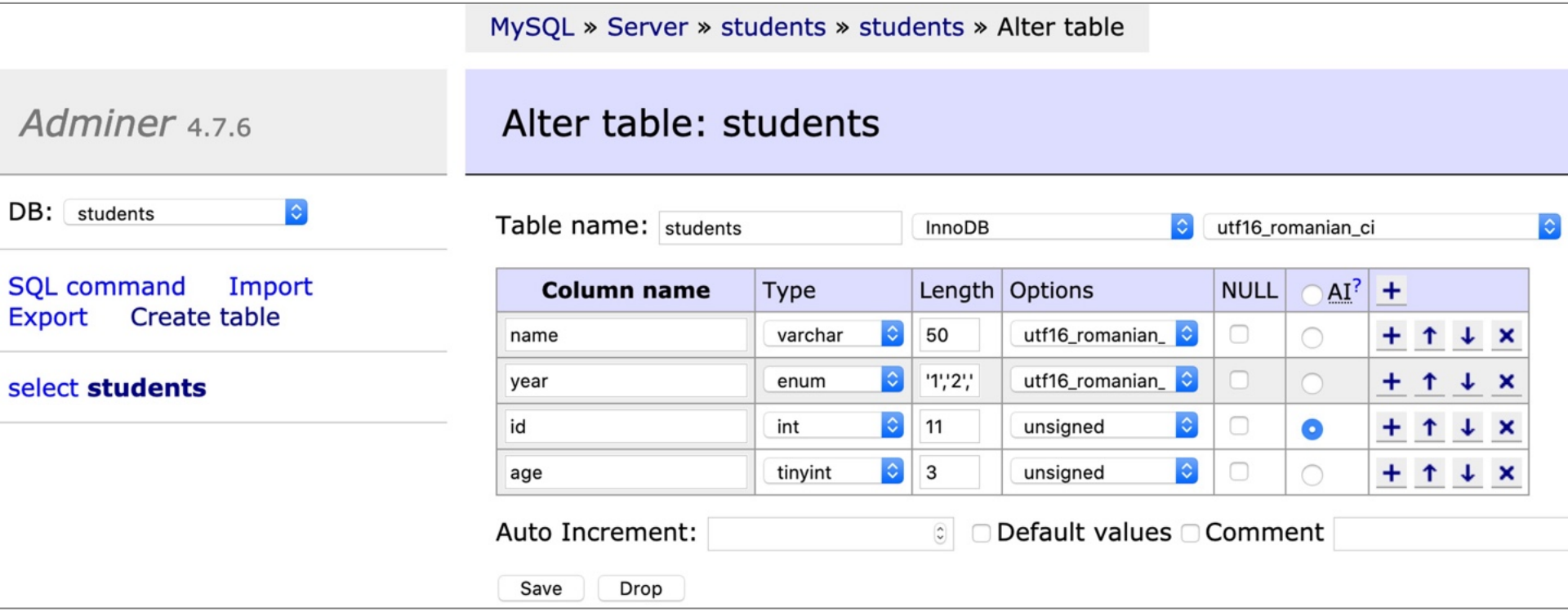
php: db – example

By using the mysql client in the command line or a Web administration application, we create the students table, having the structure:

```
+-------+---------------------+------+-----+---------+----------------+
| Field | Type                | Null | Key | Default | Extra          |
+-------+---------------------+------+-----+---------+----------------+
| name  | varchar(50)         | NO   |     | NULL    |                |
| year  | enum('1','2','3')   | NO   |     | NULL    |                |
| id    | int(11) unsigned    | NO   | PRI | NULL    | auto_increment |
| age   | tinyint(3) unsigned | NO   |     | NULL    |                |
+-------+---------------------+------+-----+---------+----------------+
```

for easy administration, use the Web tool

Adminer – **www.adminer.org**



visualising/altering the structure of the created table

alternative: phpMyAdmin – **www.phpmyadmin.net**

MySQL » Server » students » Select: students

inserting records

Select: students

Item has been inserted. 11:19:38 SQL command

```
INSERT INTO `students` (`name`, `year`, `id`, `age`)
VALUES ('Agarub Nibas Uilenroc', 2, '74', '26');
```

(0.004 s)

Edit

Select data Show structure Alter table New item

Select | Search | Sort | Limit: 50 | Text length: 100 | Action: Select

```
SELECT * FROM `students` LIMIT 50
```
(0.000 s) Edit

Modify	name	year	id	age
edit	Tuxy Pinguinnesscool	2	33	21
edit	Agarub Nibas Uilenroc	2	74	26

Whole result: 2 rows | Modify: Save | Selected (0): Edit Clone Delete | Export (2): open SQL Export

Import

MySQL » Server » students » Select: students

Select: students

inserting records

Item has been inserted. 11:19:38 SQL command

```
INSERT INTO `students` (`name`, `year`, `id`, `age`)
VALUES ('Agarub Nibas Uilenroc', 2, '74', '26');
```

(0.004 s)

Edit

Select data Show structure Alter table

Select | Search | Sort | Limit 50

```
SELECT * FROM `students` LIMIT 50
```
(0.000 s) Edit

Modify	name	year	id	age
edit	Tuxy Pinguinnesscool	2	33	21
edit	Agarub Nibas Uilenroc	2	74	26

Whole result: 2 rows | Modify: Save | Selected (0): Edit Clone

Import

MySQL » Server » students » Export

Export: students

Output	open save gzip
Format	SQL CSV, CSV; TSV
Database	Routines Events
Tables	DROP+CREATE Auto Increment Triggers
Data	INSERT

Export

Tables	Data
students	~ 2

data export

SQL command

manual execution of SQL commands

```
SELECT name, age FROM students WHERE age > 21 and year = 2
```

name	age
Agarub Nibas Uilenroc	26

1 row (0.001 s) Edit, Explain, Export

id?	select_type?	table?	partitions?	type?	possible_keys?	key?	key_len?	ref?	rows?	Extra?
1	SIMPLE	students	*NULL*	ALL	*NULL*	*NULL*	*NULL*	*NULL*	2	Using where

```
SELECT name, age FROM students WHERE age > 21 and year = 2;
```

Execute Limit rows: Stop on error Show only errors

History

query history

Edit 11:28:50 `SELECT name, age FROM students WHERE age > 21 and year = 2;`
Edit 11:28:00 `SELECT year, name, age FROM students WHERE year=2 ORDER BY age;`
Edit 11:24:19 `SELECT name, age FROM students WHERE age > 21 and year = 2;`
Edit 11:23:30 `SELECT * FROM students WHERE age > 21 and year = 2 ;`

php: db – mysqli extension

```
// a mysqli object instantiation
$mysql = new mysqli ('localhost', 'tux', 'pa$$w0Rd', 'students');
if ( mysqli_connect_errno ()) {
   die (‘Connection failed...');
}

// specifying & executing a query
if (!($res = $mysql->query ('select name, year from students'))) {
   die ('A query error occurred');
}
```

php: db – mysqli extension

```
// a mysqli object instantiation
$mysql = new mysqli ('localhost', 'tux', 'pa$$w0Rd', 'students');
if (mysqli_connect_errno()) {
   die ('Connection failed...');
}

// specifying & executing a query
if (!($res = $mysql->query('select name, year from students'))) {
   die ('A query error occurred');
}
```

The password is specified in clear text!

- **pay attention to security issues that may arise**

php: db – mysqli extension

```
// generating a numbered list of student-related data
// (HTML spaghetti code – not a recommended practice!)
echo ('<ol>');
// results will be available into an associative array
while ($row = $res->fetch_assoc ()) {
   // table column ≡ array key
   echo ('<li>Student ' . $row['name'] .
              ' is enrolled in ' . $row['year'] . ' year</li>');
}
echo ('</ol>');

// closing connection to MySQL/Maria DB server
$mysql->close ();
```

php: db

In practice, an abstraction layer is used
in order to access a storage system

DBAL – DataBase Abstraction Layer

usual approach:
PDO (PHP Data Objects)

pragmatic aspects: *(The only proper) PDO tutorial*, 2026
phpdelusions.net/pdo

```php
// connection data regarding MySQL/MariaDB database server
$host = '127.0.0.1';
$db   = 'students';
$user = 'tux';
$pass = 'pa$$w0Rd'; // important: clear text password
$charset = 'utf8';

// setting the DSN (Data Source Name)
$dsn = "mysql:host=$host;dbname=$db;charset=$charset";

// options concerning the connection manner
$opt = [
    // errors will be reported as exceptions
    PDO::ATTR_ERRMODE            => PDO::ERRMODE_EXCEPTION,
    // results will be stored by an associative array
    PDO::ATTR_DEFAULT_FETCH_MODE => PDO::FETCH_ASSOC,
    // connection is persistent
    PDO::ATTR_PERSISTENT         => true
];
```

www.php.net/manual/en/book.pdo.php

```php
// retrieve from the Web client the year of study (default: 2)
$year = $_REQUEST['year'] ? $_REQUEST['year'] : 2;

try {
    $pdo = new PDO ($dsn, $user, $pass, $opt); // instantiating a PDO object

    // preparing a parametrized SQL statement
    $sql = $pdo->prepare ('SELECT year, name, age FROM students
                           WHERE year=? ORDER BY age');

    if ($sql->execute ([$year])) {          // SQL statement can be executed?
      while ($row = $sql->fetch()) {        // getting each found record
        // ...and showing it (table column is a key of associative array)
        echo '<p>' . $row['name'] . ' is enrolled in year ' . $row['year'] . '</p>';
      }
    }
} catch (PDOException $e) {
    echo "Error: " . $e->getMessage(); // the message of occurred exception
};
```

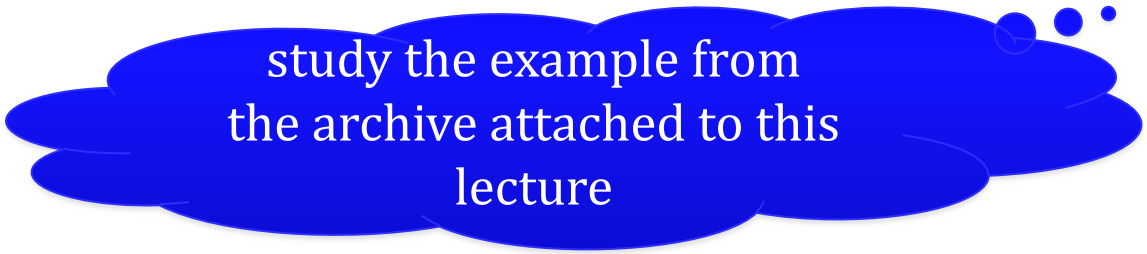
php: db – pdo extension

A possible result provided by the execution of the PHP program invoked by using a URL as
https://somewhere.info/php/pdo-test.php?year=2

Tuxy Pinguinesscool is enrolled in year 2

Grace Hopper is enrolled in year 2

Margaret Hamilton is enrolled in year 2



advanced

php: db

Usually, an ORM – Object-Relational Mapping
solution (component, library)
will be used on top of DBAL

examples:
Doctrine – **www.doctrine-project.org**
Propel – **propelorm.org**
RedBeanPHP – **redbeanphp.com**

Useful tools for Web developers?

advanced

tools: frameworks

Features:

MV* and various design patterns

database access (ORM, DAO, ActiveRecord,…)

input data validation and filtering

authentication + access control

cookie & session management

tools: frameworks

Features:

data presentation – templating

performance – i.e. caching

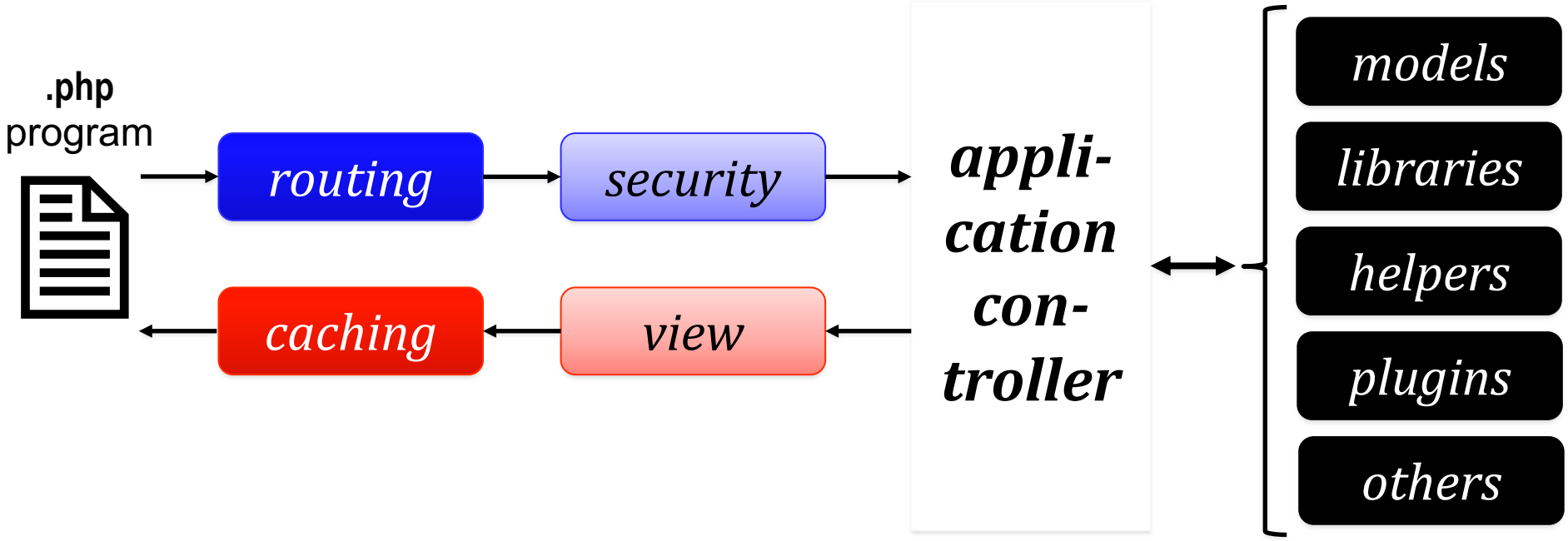
asynchronous data transfer (Ajax, WebSocket)

support for Web services and REST APIs

extensibility – e.g., user-defined modules, managed with the Composer tool

advanced

tools: frameworks

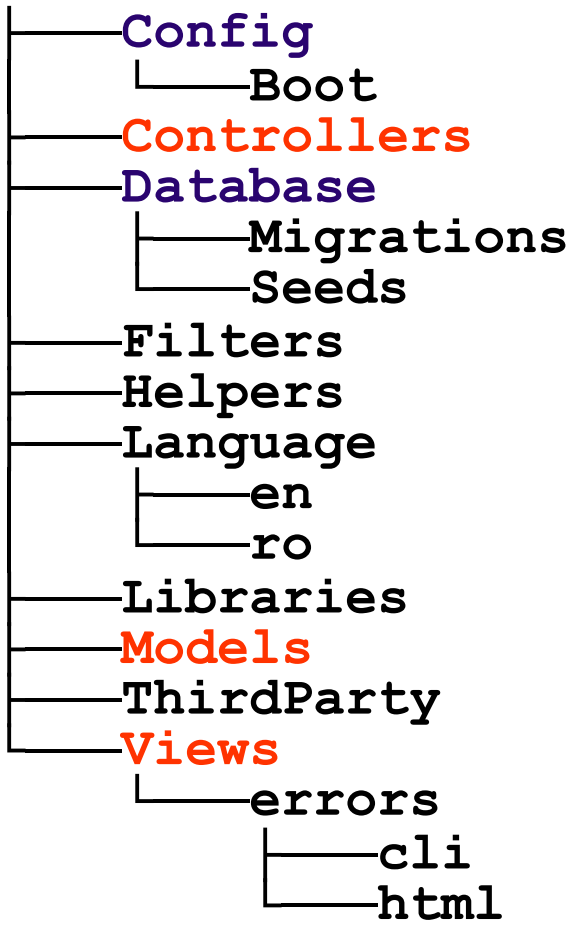


workflows performed by a framework

tools: frameworks

CakePHP – **cakephp.org**
CodeIgniter – **codeigniter.com**
FuelPHP – **fuelphp.com**
Laravel – **laravel.com**
Nette – **nette.org**
Phalcon – **phalcon.io**
PRADO – **www.pradoframework.net**
Symfony – **symfony.com**
Yii – **www.yiiframework.com**

advanced



directory structure of
a Web application developed
by using a MVC framework

CodeIgniter 4

codeigniter.com/user_guide/

tools: frameworks

Some frameworks like Laravel can benefit from the facilities offered by a virtualization platform like Docker

packaging an (Web) application and its dependencies in a virtual container that can run on any operating system installed on a host (on premises) and/or configured by a vendor – usually, in cloud

laravel.com/docs/
laracasts.com

advanced

tools: micro-frameworks

A micro-framework represents
a minimalist Web framework

advanced

tools: micro-frameworks

A micro-framework represents
a minimalist Web framework

does not include complex features

often, is focused on a single aspect regarding
Web development – e.g., creating an API, microservice,…

advanced

tools: micro-frameworks

Fat-Free Framework (F3)

fatfreeframework.com

Flight

flightphp.com

Fresh Squeezed Limonade

github.com/yesinteractive/fsl

Leaf PHP

leafphp.dev

Slim

www.slimframework.com

advanced

tools: packages

Managing dependencies
between libraries and packages

Composer

getcomposer.org

www.phptherightway.com/#dependency_management

advanced

tools: packages

Meta-data + dependencies of other (versions of) packages are specified in the file **composer.json**

placed in the / root directory of the PHP project

according to the scheme **MAJOR.MINOR.PATCH**
(semantic versioning – semver.org),
dependency versions may have constraints specified via the symbols * < <= > >= ~ ^
details at getcomposer.org/doc/articles/versions.md

advanced

```
{
   "name": “org/package",
   "description": "...",
   "homepage": "http://package.org/",
   "minimum-stability": "stable",
   "version": "1.9.74",        // current version of the package
   "license": [ "LGPL-2.1-only", "GPL-3.0-or-later" ], // license
   "authors": [ { "name": "Tuxy", "role": "Developer" } ], // author(s)
   "require": {
      "php" : "^5.5 || >=7.4",     // PHP versions required to execute the code
      “namespace/otherpackage": "~2.0", // external dependency version
      "org/package2": "*"                // another package (any vers.)
   },
   "repositories": [      // download source
      { "type": "composer", "url": "http://packages.some.where/package" } ]
}
```

Composer will be used
(other values: **git**, **vcs**, **pear**, **package**)

advanced

tools: packages

Packages are encapsulated in PHAR (PHP Archive) and can be public/private

tools: packages

Packages are encapsulated in PHAR (PHP Archive) and can be public/private

php composer.phar command

where **command** can be:

install – local/global installation including dependencies

update – package update **remove** – package deletion

search – package search

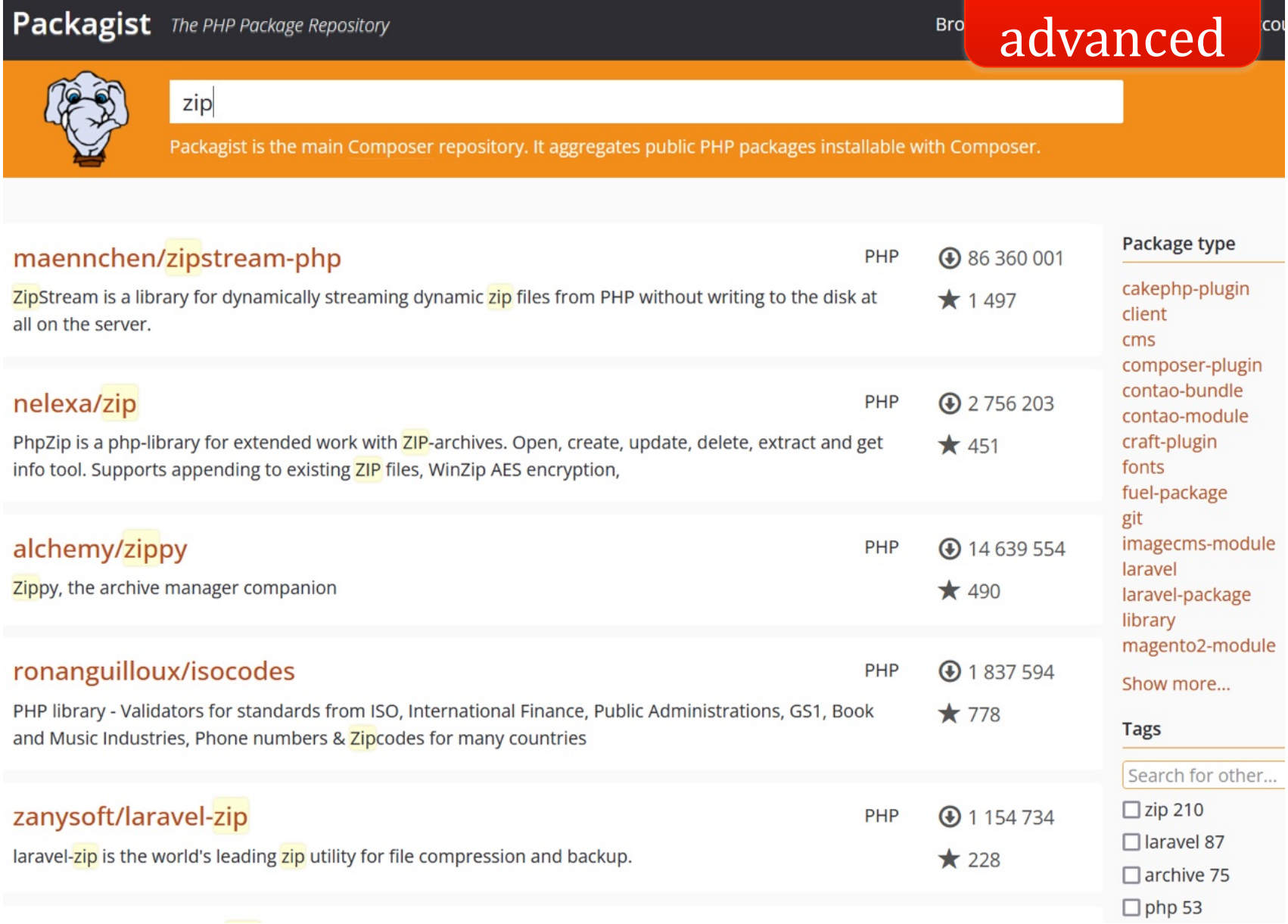
show – display all available packages

depends – display the packages that depend on a specific package

outdated – search for old packages

help – help information **diagnose** – diagnostic

...and others



public packages can be downloaded from the Web from Packagist – package repository: packagist.org

tools: packages

Dependency management
between libraries and packages

alternative – older:
PEAR (PHP Extension and Application Repository)
pear.php.net
+
extensions offered by third parties:
PECL (PHP Extension Community Library)
pecl.php.net

tools: development environments

Pre-configured Web development environments

Web server + PHP + database server(s) + tools

tools: development environments

Apache, PHP 8, MySQL/MongoDB…

AMPPS

www.ampps.com

XAMPP

www.apachefriends.org

with support for various configurations of Web systems like Drupal, Joomla!, MediaWiki, Magento, Moodle, WordPress

tools: documenting

Automatic documentation generators

Daux – **daux.io**
phpDocumentor – **www.phpdoc.org**
phpDox – **phpdox.net**

github.com/ziadoz/awesome-php#documentation

tools: documenting

Special markups in PHP comments:

@author
@category
@version
@copyright
@license
@see
@todo
@since
@deprecated

@var
@global
@method
@package
@subpackage
@param
@return
@throws
@inheritdoc

@example
@source
@uses
@used-by
@link
@internal
@property
@property-read
@property-write

docs.phpdoc.org/guide/references/phpdoc/

real example:
Wordpress

```
/**
 * WordPress User Page: Handles authentication et al.
 * @package WordPress
 */

...
/**
 * Output the login page header.
 * @since 2.1.0
 * @global string      $error     Login error message set by ...
 * @global bool | string $interim_login A login modal is being displayed...
 * @global string      $action    The action that brought the visitor to the login page.
 *
 * @param string   $title    Optional. WordPress login Page title to display
                             in the `<title>` element. Default 'Log In'.
 * @param string   $message  Optional. Message to display in header.
 * @param WP_Error $wp_error Optional. The error to pass.
 */
function login_header($title = 'Log In', $message = '', $wp_error = null ) {
  // ...
}
...
```

github.com/WordPress/WordPress/blob/master/wp-login.php

real example:
Laravel Octane

```php
/**
   * The root application instance.
   *
   * @var \Illuminate\Foundation\Application
   */
   protected $app;
...
/**
   * Handle an incoming request and send the response to the client.
   *
   * @param  \Illuminate\Http\Request  $request
   * @param  \Laravel\Octane\RequestContext  $context
   * @return void
   */
   public function handle(Request $request, RequestContext $context): void
...
```

github.com/laravel/octane/

tools: code analysis

PHP source-code static analysis

for discovering various programming bugs, verifying the conformance to various coding standards, automatic correction (fixers), determining code metrics: complexity, number of lines,…

github.com/exakat/php-static-analysis-tools

advanced

tools: code analysis

PHP Standards Recommendations – **www.php-fig.org/psr/**
written by PHP FIG (Framework Interop Group)

PSR1: *Basic Coding Standard*, PSR3: *Logger Interface*,
PSR4: *Autoloading*, PSR6: *Caching Interface*,
PSR7: *HTTP Message Interface*, PSR11: *Container Interface*,
PSR12: *Extended Coding Style Guide*,
PSR13: *Hypermedia Links*, PSR14: *Event Dispatcher*,
PSR15: *HTTP Handlers*,...

tools: phan, PHP_CodeSniffer, PHP-CS-Fixer, Rector

tools: code analysis

PSR12: Extended Coding Style Guide – example rules:

All PHP files MUST use the Unix LF (linefeed) line ending only.
*The closing **?>** tag MUST be omitted from files containing only PHP.*
Lines SHOULD NOT be longer than 80 characters.
There MUST NOT be more than one statement per line.
All PHP reserved keywords and types MUST be in lower case.
Visibility MUST be declared on all properties and methods.
There MUST be one space after the control structure keyword.

www.php-fig.org/psr/psr-12/

ERROR	TRUE, FALSE and NULL must be lowercase; expected "false" but found "FALSE"	Line 55, column 9
ERROR	Opening parenthesis of a multi-line function call must be the last content on the line	Line 55, column 26
ERROR	Expected at least 1 space before "=>"; 0 found	Line 59, column 17
ERROR	Multi-line function call not indented correctly; expected 4 spaces but found 10	Line 60, column 1
ERROR	Only one argument is allowed per line in a multi-line function call	Line 60, column 39
ERROR	Closing parenthesis of a multi-line function call must be on a line by itself	Line 60, column 43
ERROR	The closing parenthesis of a multi-line control structure must be on the line after the last expression	Line 60, column 44

reports provided by **phptools.online/php-checker**

ERROR	Whitespace found at end of line	Line 67,

advanced

tools: testing

atoum – atoum.org
Codeception – codeception.com
ParaTest (parallel testing) – github.com/brianium/paratest
Peridot – peridot-php.github.io
PHP Debugger – www.php.net/manual/en/book.phpdbg.php
PHPUnit – phpunit.de

others at github.com/ziadoz/awesome-php#testing

advanced

tools: testing

Profiling – analyzing and reporting slow-running code snippets

PhpBench – **github.com/phpbench/phpbench**
Tracy – **tracy.nette.org**
Xdebug extension for PHP – **xdebug.org/docs/profiler**

github.com/ziadoz/awesome-php#debugging-and-profiling

advanced

tools: continuous integration

CircleCI – circleci.com/docs/code-coverage/#php
Jenkins – www.jenkins.io/solutions/php/
Semaphore – semaphore.io/blog/php-microservices

github.com/ziadoz/awesome-php#continuous-integration

advanced

extensions

Hack (Facebook, from 2014)
a programming language for HHVM, extending PHP

goal: increasing the Web developer productivity

features: explicit data types (type annotations), generics, λ expressions, asynchronous programming (async), and others

hacklang.org
docs.hhvm.com

😢 major critics

many years: lack of language formal specification
now: **github.com/php/php-langspec**

inconsistency – e.g., foreach, predefined function names, multiple code writing conventions

lack of native support for Unicode (except PHP 7+)

lack of native support for multi-threading,
but possible via extensions like pthreads et al.

PHP Sadness

Function naming (underscores)

« Previous Sadness | Sadness Index | Next Sadness »

Even between similar/related functions, some use underscores between words, while others do not. Here are some examples:

get: php gettype | php get_class

str: php str_ireplace | php str_pad | php str_repeat | php str_replace | php str_shuffle | php str_split | php str_word_count | php strcasecmp | php strchr | php strcmp | php strcoll | php strcspn

encode: php base64_encode | php quoted_printable_encode | php session_encode | php rawurlencode | php urlencode | php gzencode

php: php php_uname | php php_sapi_name | php php_logo_guid | php phpinfo | php phpcredits | php phpversion

PHP Sadness – phpsadness.com

Significance: Consistency

Language consistency is very important for developer efficiency. Every inconsistent language feature means that developers have one more thing to remember, one more reason to rely on the documentation, or one more situation that breaks their focus. A consistent language lets developers create habits and expectations that work throughout the language, learn the language much more quickly, more easily locate errors, and have fewer things to keep track of at once.

USELESS ERROR REPORTING

#44 Detail in error messages

#16 Exception thrown without a stack frame

#1 Unexpected `T_PAAMAYIM_NEKUDOTAYIM`

#7 Parse error: syntax error, unexp

#54 Empty `T_ENCAPSED_AND_WHITESPAC`

INCONSISTENCY

#4 Function naming (underscores)

#15 Function naming (prefixes)

#48 Function naming (to/2)

#6 Order of arguments (array mani

#9 Order of arguments (array/string

MISSING FEATURES

#17 Standard libc process control (`fork`/`exec`/etc)

#20 (<5.4) No string escape code for ESC (ascii 27), normally `\e`

#29 Can't decode php-session-format serialization outside a session

#49 (<5.5) No support for `finally`-type error handling

OBJECT-ORIENTED SYSTEM ISSUES

#8 Implementing all the right methods (array) still doesn't work in array functions

#18 (intended) Static variables in methods are bound to all instances of the class

#24 Seemingly redundant reflection methods with no documentation

😢 major critics

In PHP, we can – easily/involuntarily? –
create applications that “adopt”
the Spaghetti Code anti-pattern

study Unix Sheikh et al., *PHP The Wrong Way*, 2025
phpthewrongway.com

```
<h1>My Users <a class="btn btn-primary" href="new_user.php" role="button">New User</a></h1>
<table class="table table-condensed">
    <tr>
        <th>Id</th>
        <th>Name</th>
        <th>Age</th>
        <th>Email</th>
        <th>Action</th>
    </tr>
    <?php
    // Get all users
    $stmt = $dbh->prepare("SELECT * FROM users");
    $stmt->setFetchMode(PDO::FETCH_ASSOC);
    if ($stmt->execute()) {
        while ($row = $stmt->fetch()) {
            ?>
            <tr>
                <td><?php echo $row['id']; ?></td>
                <td><?php echo $row['name']; ?></td>
                <td><?php echo $row['age']; ?></td>
                <td><?php echo $row['email']; ?></td>
                <td>
                    <div class="btn-group" role="group" aria-label="...">
                        <a href="edit.php?id=<?php echo $row['id']; ?>" class="btn btn-default btn-sm">Edit</a>
                        <a href="index.php?delete=<?php echo $row['id']; ?>" class="btn btn-default btn-sm">Delete</a>
                    </div>
                </td>
            </tr>
            <?php
```

HTML Web interface

PHP for data access and processing

PHP for data presentation

github.com/hanafiah/masterphp/tree/master/spaghetti

A few case studies of Web applications developed in PHP?

WIKIPEDIA
The Free Encyclopedia

Search Wikipedia | Search

World Wide Web

150 languages

Contents hide

(Top)
History
Nomenclature
Function
HTML
Linking
WWW prefix
Scheme specifiers
Pages
Static page
Dynamic pages
Website
Browser
Server
Optical Networking
Cookie
Search engine
Deep web
Caching
Security
Privacy
Standards
Accessibility
Internationalisation
See also
References

Article Talk | Read View source View history Tools

From Wikipedia, the free encyclopedia

This article is about the global system of pages accessed via HTTP. For the worldwide computer network, see Internet. For the web browser, see WorldWideWeb.
"WWW" and "The Web" redirect here. For other uses, see WWW (disambiguation) and The Web (disambiguation).

The **World Wide Web** (**WWW** or simply **the Web**) is an information system tha content sharing over the Internet through user-friendly ways meant to appeal to beyond IT specialists and hobbyists.[1] It allows documents and other web reso be accessed over the Internet according to specific rules of the Hypertext Trans Protocol (HTTP).[2]

A **web application** is application software that is created with web technologies and runs via a web browser. Web applications emerged during the late 1990s and allowed for the server to dynamically build a response to the request, in contrast to static web pages.

The Web was invented by English computer scientist Tim Berners-Lee while at 1989 and opened to the public in 1993. It was conceived as a "universal linked information system".[3][4][5] Documents and other media content are made avail accessed by programs such as web browsers. Servers and resources on the W character strings called uniform resource locators (URLs).

The original and still very common document type is a web page formatted in H language supports plain text, images, embedded video and audio contents, and interaction. The HTML language also supports hyperlinks (embedded URLs) wh Web navigation, or web surfing, is the common practice of following such hyperlinks across multiple websites. Web applications are web pages that function as application software. The information in the Web is transferred across the Internet using HTTP. Multiple web resources with a common theme and usually a common domain name make up a website. A single web server may provide multiple websites, while some websites, especially the most popular ones, may be provided by multiple servers. Website content is provided by a myriad of companies, organizations, government agencies, and individual users; and comprises an enormous amount of educational, entertainment, commercial, and government information.

The Web has become the world's dominant information systems platform.[6][7][8][9] It is the primary tool that billions of people worldwide use to interact with the Internet.[2]

founders: Jimmy Wales & Larry Sanger (2001)

case study: wikipedia

Aim: providing open content by using a group of collaborative Web applications – wikis

case study: wikipedia

Aim: providing open content by using a group of collaborative Web applications – wikis

Wikipedia Foundation
also maintains Wikipedia, Wiktionary, Wikibooks, Wikiquote, Wikivoyage, Wikisource, Wikimedia Commons, Wikispecies, Wikinews, Wikiversity, Wikidata
en.wikipedia.org/wiki/Wikimedia_Foundation

case study: wikipedia

Complex version of the LAMP technology stack

MediaWiki - *wiki* system used for all services
implemented in PHP (8.2+ versions) + JavaScript
MariaDB (main storage solution;
alternatives: SQLite, PostgreSQL)
ImageMagick, DjVu, TeX, rsvg, ploticus, etc.
(for graphical content processing within MediaWiki)
Apache HTTP Server + NGINX (Web servers)

also provides an API to be used by Web developers:
www.mediawiki.org/wiki/API:Main_page

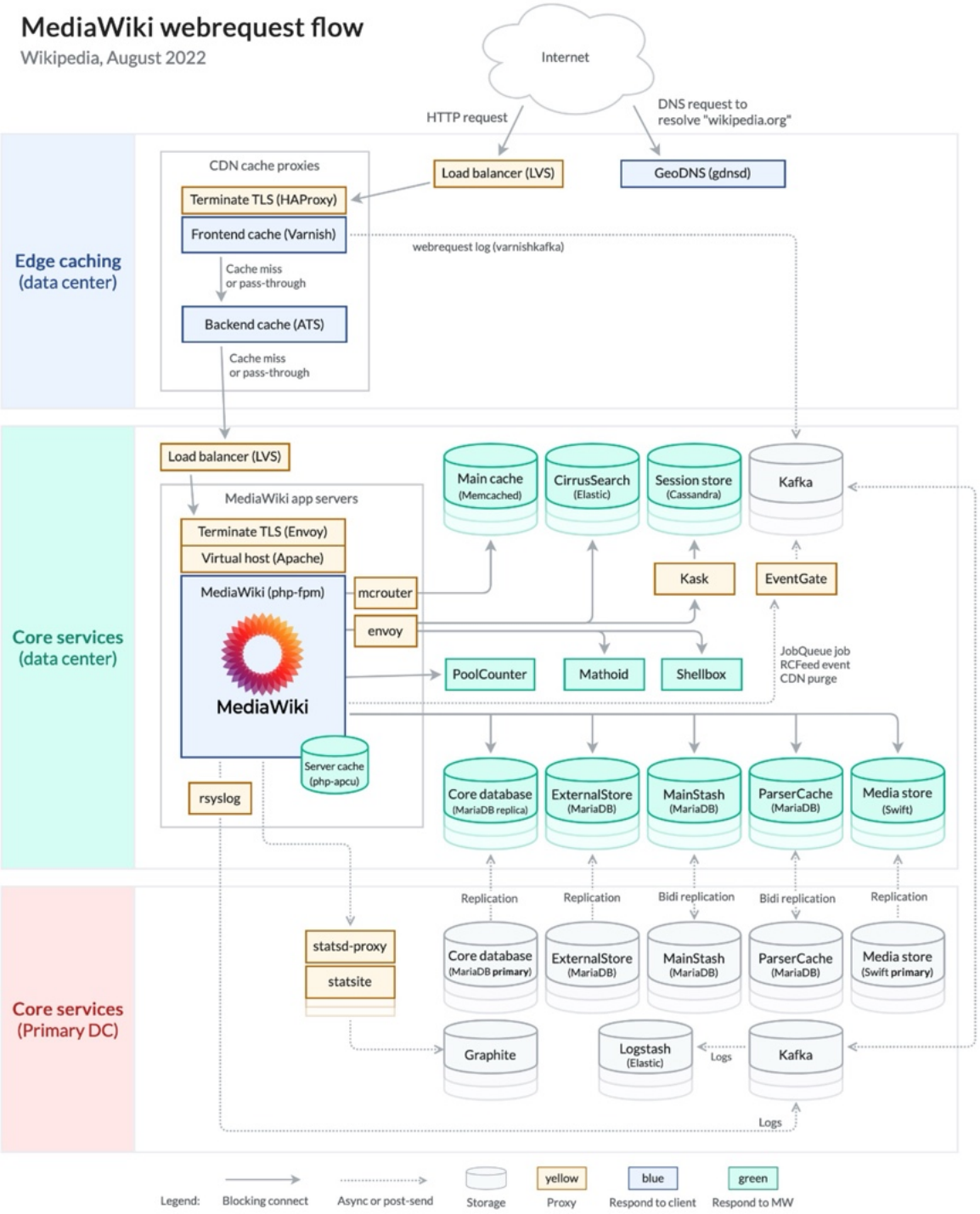
advanced

case study: wikipedia

PHP-FPM (FastCGI Process Manager for PHP)
Apache Traffic Server + Varnish (proxy + caching)
Memcached (caching for database queries)
OpenSearch (textual search –Java implementation)
gdnsd (C++ solution for DNS)
Apache Kafka (event streaming – Java + Scala)
Linux Virtual Server (load balancing); PyBal (local monit.)
Debian GNU/Linux (OS)
Graphana+Icinga (monitoring): **www.wikimediastatus.net**
Phabricator (bug tracking)

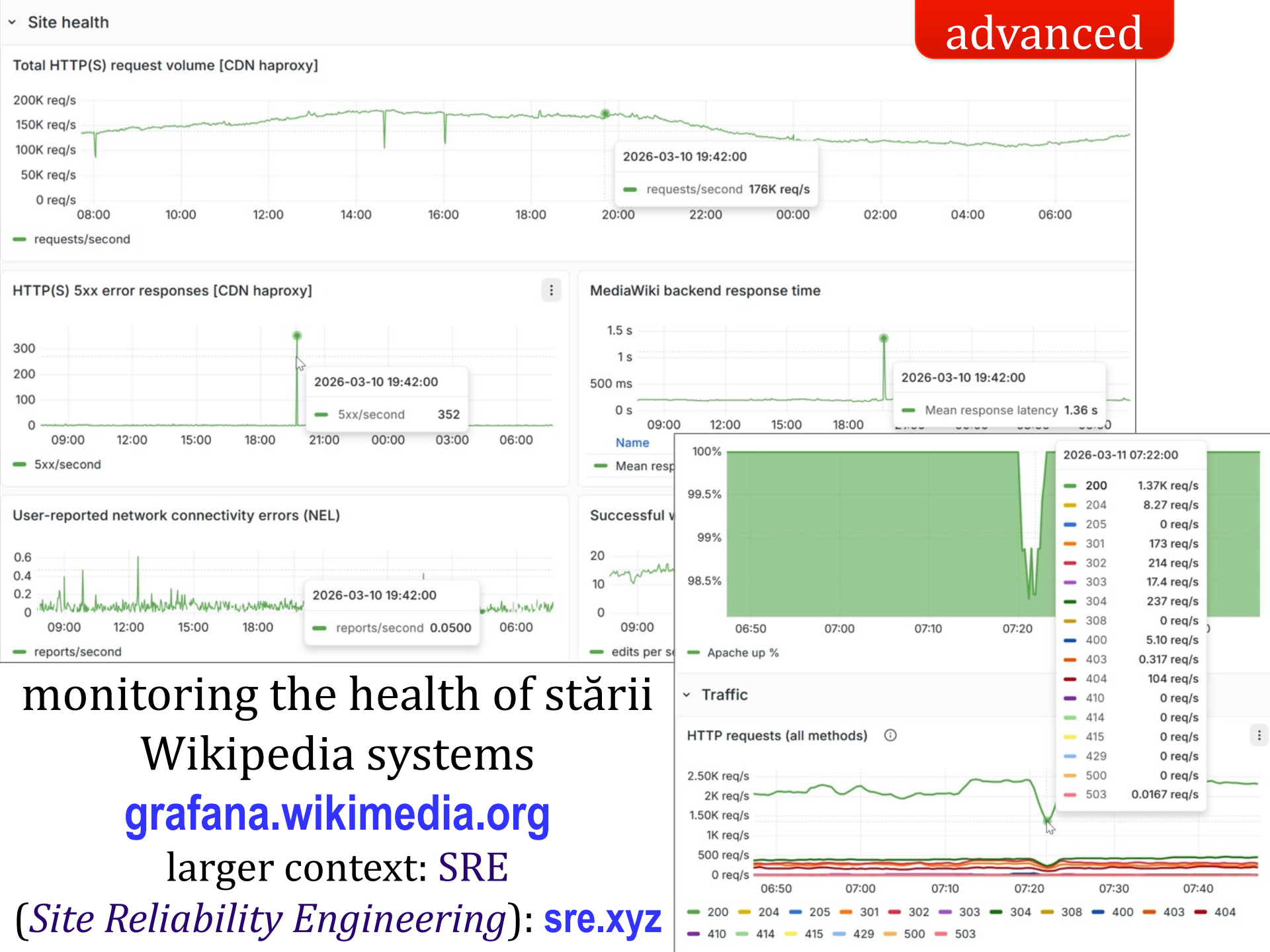
meta.wikimedia.org/wiki/Wikimedia_servers

advanced



main hosting:
USA, Virginia
secondary hosting:
USA, Texas

replication (caching) in
Europe: Amsterdam
Europe: Marseille
USA: San Francisco
Asia: Singapore
South America: São Paulo



monitoring the health of stării Wikipedia systems

grafana.wikimedia.org

larger context: SRE (*Site Reliability Engineering*): **sre.xyz**

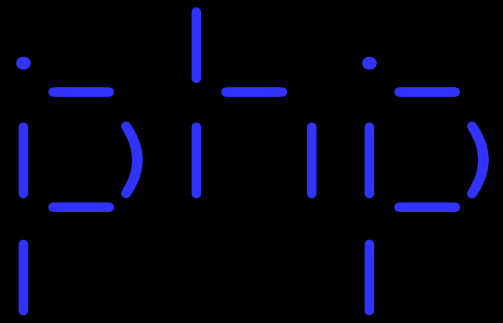
case studies

Various Web sites use Content Management Systems (CMS) developed in PHP

general:
Drupal, Joomla, WordPress, etc.
wiki:
DokuWiki, MediaWiki, pmWiki, etc.
for e-commerce:
Magento, OpenCart, PrestaShop,…

summary

PHP overview



characteristics, features, tools, examples

```xml
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE html>
<!-- pentru redarea datelor, se recurge la o foaie de stiluri CSS -->
<?xml-stylesheet type="text/css" href="game.css" ?>
<game>
    <title>Angry Profs</title>
    <platform type="tablet">Android</platform>
    <platform min-version="9" type="tablet">iOS</platform>
    <platform min-version="10">Windows</platform>
    <url>...</url>
    <player>
        <identity>
            <first-name>Sabin</first-name>
            <last-name>Buraga</last-name>
            <!-- ... -->
        </identity>
        <points>30374</points>
    </player>
</game>
```

next episode:

a data model for Web: XML family